

## Meridian Casualty: Hiring Is Broken. Can You Fix It?

Econ Week 2026 case competition, Round 1 written analysis. Team: Ethan Friedman, Austin Glick, Danny Shin and Sam Stark. Every number cites an appendix exhibit with sample sizes, standard errors and confidence intervals.

### **Recommendation: shift the cost of screening back onto applicants. Re-time the situational section, block pasting, and flag fast answers**

Meridian should move the situational section (the essay, in the data) on its new applicant tracking system (ATS) back to the timed, no-paste format its seven legacy centers still use, and show recruiters the minutes each applicant spent in the section. That is option (b). Against the status quo it lowers the cost of one retained employee by **\$713 to \$1,556** when operational quality is left out, and by **\$2,548 to \$3,392** when it is monetized (E19). At Meridian's current new-ATS hiring pace that is **\$4.9 million to \$10.6 million a year** before quality and before the vendor's implementation charge, which the data do not contain (E19). The difference-in-differences gain alone gives the low end, and the seven legacy centers' current results give the high end (E19 rows b-low to b-high). The quality figures also rest on the legacy centers' lower reopen rate, whose difference-in-differences interaction on its own does not clear zero (E02b), so we defend the excluding-quality range.

Since 2023 the essay no longer predicts who stays on the new system, it still carries part of that signal on the legacy system, and the legacy centers retain more hires today (E12, E12b, E12d, E01b, E17). Only option (b) has a retention effect we can observe in Meridian's own centers rather than assume.

Seven small legacy centers carry most of the weight here, and that is the hole in our analysis. They sit in the Southeast and Mid-Atlantic, and their own essay signal is eroding. Matching them to new-ATS centers on size and region does not shrink their advantage, but seven centers can only bound it (E15f, E07b, E12e). Their result therefore caps what (b) can deliver, and we report the range rather than its midpoint.

| <b>Per retained employee, 2025 rates</b> | <b>Status quo</b> | <b>Option (b)</b>    | <b>Saving</b>                     |
|------------------------------------------|-------------------|----------------------|-----------------------------------|
| Excluding operational quality            | \$23,377          | \$21,821 to \$22,665 | \$713 to \$1,556                  |
| Including operational quality            | \$40,908          | \$37,516 to \$38,360 | \$2,548 to \$3,392                |
| Annual, 6,830 retained seats             |                   |                      | \$4.9M to \$10.6M (excl. quality) |

Source: E17, E19. Six-month retention  $r = 0.820$  at the status quo and  $0.843$  to  $0.857$  under (b).

### **(a) The process still measures the right things, but it keeps full weight on an essay signal that has vanished, and recruiters have half the time to read it**

**What works.** Four screen inputs still predict who lasts six months, and each predicts better in the AI era than before it. Only the pre-2023 legacy interview slope fails to clear zero (E11b). Holding the other inputs fixed, an adjuster license raises six-month retention by 1.6 points on the legacy system before 2023 and 4.9 points on the new system after 2023. A referral goes from 1.6 to 3.2 points, the resume score from 1.0 to 1.9 points per unit, and the structured interview from 0.9 to 2.7 points per unit (E11, E11b, E14). Applicant volume works. Openings fill in about a month and each draws over a hundred applications (E10, E10b).

**First failure: the screen has not adapted.** The recruiter screen score is a fixed formula. The essay's weight inside it is 0.250 in every ATS-and-era cell, with an R-squared of 0.82 to 0.86, exactly what it was when the essay worked (E11c). Meanwhile the essay's own predictive slope on retention has gone from 0.8 points per rubric point on the legacy system before 2023 to zero on the new system after 2023 (E11b). Meridian is still paying a quarter of its screen for information that is no longer there. Part of this failure is fixable at no cost and without touching the application: move that weight from the essay to license and referral, which still predict retention in this cell (E11b). We do not cost this step, because it only pays off if it changes who gets hired, and we cannot observe outcomes for applicants the current formula did not select. It is still the first thing Meridian should try, alongside anything below.

**Second failure: recruiters get less time per application every year, and the new-ATS centers get the least.** Applications per opening rose from 150 in 2021 to 217 in 2025 while recruiter minutes per application fell from 6.2 to 3.0 and days to fill rose from 22 to 30 (all centers pooled, E10e, and by ATS, E10b). Each extra applicant per opening costs 0.023 minutes of review per application and lengthens the time to fill (E10f, E10c). The new ATS itself draws 348 more applications per center-month than the legacy system in the AI era, most likely because an untimed section that accepts pasted text is easier to finish (E10d). So the new-ATS centers get both the least informative essays and the least time to read them.

**Whitfield sees the consequence.** Six-month retention on the new ATS fell 2.3 points more than on the legacy system once the AI era began (difference-in-differences  $-0.023$ , 95% CI  $-0.042$  to  $-0.004$ , 41,677 hires, E01, E01b). Reopen rates on the new ATS are 0.14 to 0.25 percentage points higher in the AI era. The interaction term gives 0.14 and the raw AI-era gap 0.25, and only the 0.25 clears zero (E02, E02b). Separations rose across voluntary, performance and attendance exits, with no single reason carrying the increase (E03, E03b, E03c). That fits a screen that admits people who neither stick nor perform, since a labor-market story would show mainly as voluntary exits. Her "best-run centers are worst" describes the pilot wave. The largest centers migrated first in 2021, held retention at 0.878 through 2022, then fell to 0.830 by 2024-25, while never-migrated centers moved from 0.889 to 0.857. The decline appears in every size tercile (E04, E05).

**(b) The situational section no longer screens on the new system, and we know because its slope on outcomes collapsed in one cell and nowhere else**

**The comparison.** We estimate the slope of the essay rubric score on the six-month outcome within each of four cells: legacy or new ATS at the application month, crossed with application before or after January 2023. **On whom.** Two samples. Permanent hires (41,677 with an observed outcome, after dropping 4,589 censored hires, E00) were selected on the essay, so every slope in that sample is attenuated. Flex placements (28,672) applied to Meridian but were placed by the staffing partner on certification, geography and availability. The essay played no part, so their slope comes out cleaner. Both use center and month fixed effects with standard errors clustered by center.

**The result.** Among hires, a rubric point raised retention by 0.87 points on the legacy system before 2023 (95% CI 0.69 to 1.05) and 1.07 points on the new system before 2023 (0.74 to 1.40). On the new system after 2023 the slope is  $-0.26$  points ( $-0.49$  to  $-0.02$ ), so a higher essay score now predicts slightly worse retention (E12b), and the same reversal appears on hires' reopen rates (E12c). On the legacy system after 2023 it is 0.48 points (0.14 to 0.82), roughly half its old value (E12b). Flex placements tell the same story on reopen rates. Each rubric point lowered the reopen rate by 0.18 points on legacy and 0.21 on the new system before 2023, by 0.12 on legacy after 2023, and by only 0.07 on the new system after 2023 (E12d). Without a regression, E12 shows a 24-30 essay retaining better than an 18-23 essay in three cells and worse in the fourth, 0.829 against 0.853. Combining the two samples, the new ATS in the AI era keeps **0% to 39%** of the essay's pre-2023 predictive power and the legacy ATS keeps **55% to 68%** (E15e). The two samples bracket the truth, because hires understate the loss and flex placements do not.

**Time spent in the section explains the collapse.** Among hires who spent 15 minutes or more writing, the essay still predicts retention in every cell, including the new system in the AI era (0.41 points per rubric point, 0.20 to 0.61). Among those who spent under 15 minutes on the new system after 2023, a higher score predicts worse retention ( $-1.64$  points,  $-2.06$  to  $-1.22$ ) (E07, E13, E13b). On the legacy system before 2023 a minute in the section added 0.054 rubric points. On the new system after 2023 a minute subtracts 0.021 (E13c). A high score written in a few minutes looks like a prepared answer pasted into the box. Meridian already records the minutes and does not use them.

**Placebo.** If something center-wide had changed, every input would have lost power. None did. The resume slope went from 0.9 to 2.3 points, license from 2.0 to 6.1, prior claims years from 0.4 to 0.8, interview from -0.5 to 3.2, over the same cells in which the essay went from 0.87 to -0.26 (E14).

**Timing.** Essay scores rise from 2023 in every new-ATS center regardless of when it migrated, including the 2021 pilots, so the break comes from the AI era meeting an untimed field rather than from the software itself (E06). A grid search over break dates finds a ramp rather than a jump, so any cutoff in 2023 tells the same story (E06b). With center, month and applicant controls, the new ATS raised scores by 0.89 rubric points before 2023 and a further 0.79 (0.58 to 1.00) once AI writing arrived (E06c), while objective attributes drifted slightly down (license -2.6 points, prior claims experience -2.4, referral -1.1, E08, E08b) and the flex pool's reopen rate did not move (E09, E09b), so the scores rose while the pool did not improve.

**Identification.** Before 2023 the pilot centers' essay slope matched the never-migrated centers' (0.010 against 0.008 per rubric point, E15g), so migration did not select on essay validity. The event study shows no pre-trend, then a 0.029 widening after migration, which matches the difference-in-differences estimate from a different design (E15, E15b). Unemployment and wages move together across ATS groups (E16, E16b). The essay's standard deviation among hires is 2.85 to 2.98 in every cell, so the zero slope is not range restriction. The share of high scorers changed instead. In the AI era 23% of new-ATS applicants score 24-30, against 8% of legacy applicants before 2023 (E15c). With only 40 centers, a wild-cluster bootstrap on the new-ATS AI-era minus legacy pre-2023 slope difference gives p below 0.01 in both samples (E15d).

**What must be true** for this to mean the essay stopped screening. The staffing partner does not place on the essay (stated in the case). Nothing unobserved changed at migrating centers in 2023 that also changed essay validity (E15 and the pre-period parity above). And the 2023 cutoff is substantively right (E06b).

**What we could not determine.** Whether any individual used AI. What the 93% to 95% of applications never reviewed look like (E10e). And how far the legacy system's own erosion continues, since its sub-10-minute share doubled from 6% to 12% and its bootstrap p on erosion is 0.08 for hires and 0.003 for flex (E07b, E15d), so a timed field slows the leak without stopping it.

### **(c) Only option (b) has an observed retention effect, and it wins under every bound**

We implement Meridian's worksheet exactly and reproduce the worked example (E17a). We cost the status quo on new-ATS hires starting in 2024-25:  $r = 0.820$ , 28.9 days to fill, an 8.9% reopen rate, and a cost per retained employee of \$23,377 excluding quality and \$40,908 including it (E17). Three counterfactuals in which the essay still screened put the cost of the lost information at \$713 to \$1,830 per retained employee excluding quality and \$2,548 to \$4,334 including it (E17). We show both because the quality term roughly doubles every gap and can change the ranking.

E19 costs the four options against the status quo per retained employee. **(a) Do nothing** gains nothing. Retention did not move from 2024 to 2025 (+0.000), so extending that trend shifts the cost by \$23, which is noise. **(b) Applicants bear it** saves \$713 to \$1,556 excluding quality, from the difference-in-differences gain alone up to the legacy centers' 2024-25 retention, reopen rate and shorter fill time. The size-and-region matched estimate, \$1,379, sits inside that range (E15f, E19). **(c) Meridian bears it**, a structured interview at 15 extra recruiter minutes per reviewed application (\$204 per hire) and 3 more days to fill, costs \$448 more excluding quality and \$469 less including it, if it recovers half the gap. We assume that recovery, since no center has run the longer interview. **(d) Bear it after hiring**, a \$500 six-month bonus, costs \$305 more either way, because separations are front-loaded. Of those who leave, 47% are gone by month two and only 33% leave in months four to six, before a probation review can act (E18, E18b). Recruiter screening, which the prompt puts inside  $C\_hire$ , is about \$740 per hire (E17b). We leave it out of every scenario because it would fall under (b), which draws fewer applications per opening (E10b, E10d), so its omission understates (b)'s saving. Option (c) is charged its extra minutes.

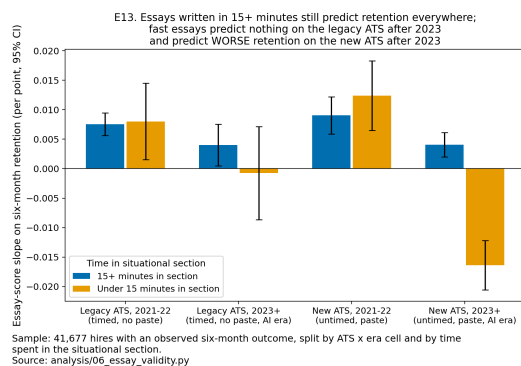
**Sensitivity.** The prompt's warning holds. Option (b) trades fewer separations for possibly more vacancy days. At \$219 per vacancy day, (b)'s saving is erased by 2.7 extra days to fill if it only buys the difference-in-differences gain, 4.4 days at the legacy-observed retention, 5.5 at the matched estimate, and 7.4 at the pre-AI level. Including quality, the break-evens are 9.8 to 14.8 days (E20, E20b, E20c). In the data the legacy centers fill faster (27.3 against 28.9 days in the 2024-25 cohorts, E17), and the matched gap is still about 0.7 days in their favor (E15f). The "fewer applicants, longer fill" fear does not show up, though those centers are small.

**What (b) costs applicants.** Whitfield is right that a timed, no-paste section is harder to complete. Start with drop-off. The legacy centers drew 183 applications per opening against 224 on the new system in 2025 (E10b), so a timed field could cost about a fifth of the queue. That is already priced in, because (b) borrows the legacy centers' retention, reopen and fill figures (E19), which they achieve with the smaller queue, and sourcing cost is charged per opening rather than per applicant (E17a). Then fairness. A timer penalizes slow writers, including applicants who need accommodation, so the vendor ask should include extended time on request and the pilot should report completion rates by applicant group. We could not measure either cost, and drop-off under a timed section is the fifth thing we watch for in the Round 2 packet.

**We would recommend something else if:**

- **(c) instead**, if a timed section pushed days to fill up by more than the break-evens above, or if a pilot showed a structured interview recovers more than 89% of the legacy retention gap at \$204 per hire (E20c).
- **(d) instead**, if separations were concentrated in months four to six rather than the first two (E18).
- **Do nothing**, if the Round 2 packet showed the legacy centers' advantage disappears once matched on size and region (on our data it does not, E15f), or that the vendor cannot re-time the section for less than the annual saving in E19.

**Implementation, in three steps.** First, at no cost, show recruiters the minutes-in-section field Meridian already collects and flag answers under 15 minutes (E13b). Second, ask the vendor for a fixed quote to make the section timed and paste-disabled on the new ATS. The annual saving in E19 is the most that change can cost and still pay. Third, pilot the timed section for one hiring quarter in the centers that migrated in 2024, with two stop rules: days to fill rising by more than the break-even in E20c, or applications per opening falling below the legacy centers' level (E10b). The nine 2024-wave centers make about 425 hires a quarter (E00), enough to detect a five-point retention gain but not the 2.3-point lower bound, so use the first quarter to check the stop rules and read retention against the difference-in-differences bound over two to three quarters before rolling out. Five things in the Round 2 packet could change this call: the vendor's cost and feasibility, the matched comparison, a labor-market shock, a revised reopened-claim cost, and applicant drop-off.



## Appendix: exhibits

Exhibits appear in register order (E00 to E20), which follows the narrative's sections. Every figure carries its title, axis units, sample statement and source script inside the image; every table is reproduced from outputs/tables/ exactly as the analysis scripts wrote it, with its sample, fixed effects, controls and standard-error type in the notes beneath. Regressions are linear probability models for binary outcomes; standard errors are clustered by center unless a note says otherwise.

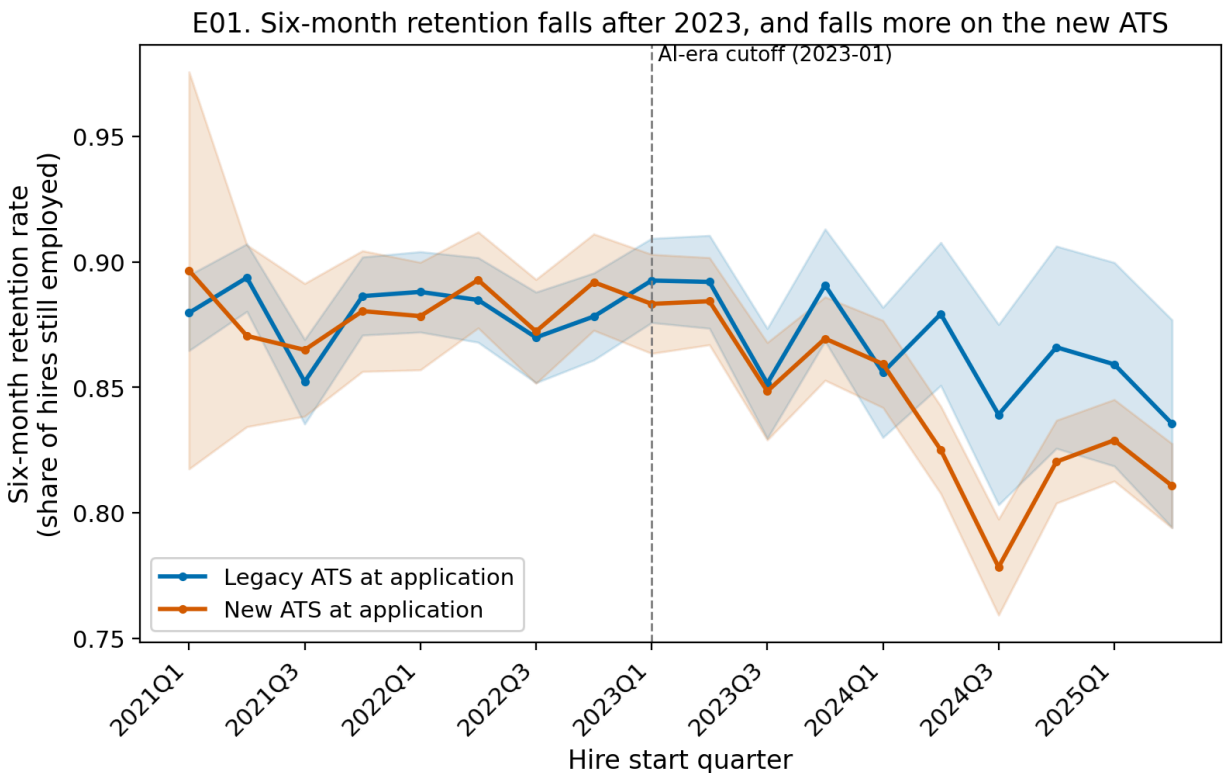
| Exhibit | Title                                                                                                                                                                  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E00     | Sample construction and data-integrity checks                                                                                                                          |
| E01     | Retention by quarter ATS (figure)                                                                                                                                      |
| E01b    | Retention DiD: new ATS x post-AI-era interaction                                                                                                                       |
| E02     | Reopen by quarter ATS (figure)                                                                                                                                         |
| E02b    | Reopen-rate DiD: new ATS x post-AI-era interaction                                                                                                                     |
| E03     | Separation reason by year (figure)                                                                                                                                     |
| E03b    | Separation reason share by start year (%)                                                                                                                              |
| E03c    | Separation-reason DiD: new ATS x post-AI-era interaction, by reason                                                                                                    |
| E04     | Retention by wave year (figure)                                                                                                                                        |
| E05     | Retention by wave size tercile (figure)                                                                                                                                |
| E06     | Essay score by ATS monthly (figure)                                                                                                                                    |
| E06b    | Grid search for the AI-era break date in new-ATS essay scores                                                                                                          |
| E06c    | Essay-score inflation: new ATS x post-AI-era interaction                                                                                                               |
| E07     | Essay minutes distribution (figure)                                                                                                                                    |
| E07b    | Mean essay score and share of essays under 10 minutes, by year and ATS                                                                                                 |
| E08     | Objective attributes over time (figure)                                                                                                                                |
| E08b    | Objective applicant attributes: new ATS x post-AI-era DiD                                                                                                              |
| E09     | Flex reopen over time (figure)                                                                                                                                         |
| E09b    | Flex reopen rate: new ATS x post-AI-era interaction                                                                                                                    |
| E10     | Recruiter capacity panel (figure)                                                                                                                                      |
| E10b    | Recruiter-capacity metrics by year and ATS                                                                                                                             |
| E10c    | Drivers of days-to-fill                                                                                                                                                |
| E10d    | Applications received per center-month: new ATS x AI-era DiD                                                                                                           |
| E10e    | Recruiter-capacity metrics by year, all 40 centers pooled                                                                                                              |
| E10f    | Recruiter minutes per application vs applications per opening                                                                                                          |
| E11     | Screen component coefficients by regime (figure)                                                                                                                       |
| E11b    | Predictors of six-month retention, by regime                                                                                                                           |
| E11c    | How much weight recruiters place on the essay score, by regime                                                                                                         |
| E12     | Retention by essay bin regime (figure)                                                                                                                                 |
| E12b    | Essay-score slope on six-month retention, by regime (hires)                                                                                                            |
| E12c    | Essay-score slope on six-month reopen rate, by regime (hires)                                                                                                          |
| E12d    | Essay-score slope on reopen rate, by regime (flex placements -- NOT selected on the essay by Meridian)                                                                 |
| E12e    | Spec C: essay slope on retention in the 12 months before vs after a center's own ATS migration (2023/24 waves) vs never-migrated centers over the same calendar window |
| E13     | Essay slope by speed regime (figure)                                                                                                                                   |
| E13b    | Essay-score slope on retention, by regime and by whether the applicant spent under 15 minutes in the situational section                                               |
| E13c    | Essay score per minute spent in the section, by regime                                                                                                                 |
| E14     | Placebo test: slope of other screen inputs on six-month retention, by regime                                                                                           |
| E15     | Migration event study (figure)                                                                                                                                         |
| E15b    | Event-study means: retention before/after migration vs never-migrated                                                                                                  |
| E15c    | Selection on the essay into the hire sample, by regime                                                                                                                 |
| E15d    | Small-cluster inference: wild-cluster bootstrap on the headline essay-slope differences                                                                                |
| E15e    | Share of the legacy pre-2023 essay signal retained, by cell: hires sample vs flex sample                                                                               |
| E15f    | Matched legacy comparison (figure with table)                                                                                                                          |
| E15g    | Pre-2023 essay slope on retention: pilot centers vs never-migrated centers                                                                                             |
| E16     | Labor market by ATS (figure)                                                                                                                                           |
| E16b    | Labor-market conditions by year and ATS                                                                                                                                |
| E17     | Cost per retained employee: status quo vs counterfactuals where the essay still screened                                                                               |
| E17a    | Worksheet inputs: 2025 average rates across 40 centers vs the prompt's worked example                                                                                  |

| Exhibit | Title                                                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------------|
| E17b    | Recruiter screening time per hire (not in the prompt's C_hire; shown for option c)                                             |
| E18     | Separation hazard by month (figure)                                                                                            |
| E18b    | Separations by month on the job and reason (counts)                                                                            |
| E19     | Cost per retained employee by option, vs status quo                                                                            |
| E20     | Sensitivity option b (figure)                                                                                                  |
| E20b    | Sensitivity table: cost per retained employee under option (b), \$ excl. quality, by r (rows) and extra days to fill (columns) |
| E20c    | Break-even: extra vacancy days that erase option (b)'s saving; recovery share option (c) needs                                 |

## E00. Sample construction and data-integrity checks

| check                                                                               | value                                             | expected      |
|-------------------------------------------------------------------------------------|---------------------------------------------------|---------------|
| applications.csv rows                                                               | 529,302                                           | 529,302       |
| hires.csv rows                                                                      | 46,266                                            | 46,266        |
| flex_placements.csv rows                                                            | 28,672                                            | 28,672        |
| center_month.csv rows                                                               | 2,400                                             | 2,400         |
| ats_migration.csv rows                                                              | 40                                                | 40            |
| hires_joined rows (== hires.csv)                                                    | 46,266                                            | 46,266        |
| flex_joined rows (== flex_placements.csv)                                           | 28,672                                            | 28,672        |
| applications with outcome=hired                                                     | 46,266                                            | 46,266        |
| applications with outcome=flex_placement                                            | 28,672                                            | 28,672        |
| hires censored (no 6mo outcome)                                                     | 4,589                                             | expect ~4589  |
| hires with observed 6mo outcome (analysis sample)                                   | 41,677                                            | expect ~41677 |
| coaching_hours null total                                                           | 4,161                                             | n/a           |
| coaching_hours null AND censored                                                    | 571                                               | n/a           |
| coaching_hours null AND NOT censored (true missing)                                 | 3,590                                             | n/a           |
| hires: ATS at start != ATS at application (use application-month ATS)               | 530                                               | expect ~530   |
| application-to-start lag distribution (months)                                      | 0 mo: 11,169; 1 mo: 31,475; 2 mo: 3,609; 3 mo: 13 | n/a           |
| hires_obs by regime (cutoff 2023-01)                                                | legacy: 19,366; new_pre: 5,920; new_post: 16,391  | n/a           |
| flex by regime (cutoff 2023-01)                                                     | legacy: 14,620; new_pre: 2,346; new_post: 11,706  | n/a           |
| centers never migrated to new ATS                                                   | 7                                                 | expect 7      |
| hires starting in 2025 at the 9 centers that migrated in 2024 (all, incl. censored) | 1,698                                             | n/a           |
| = hires per quarter at those centers                                                | 424                                               | n/a           |

*Sample:* All five raw files, 2021-01 to 2025-12, 40 centers. *Notes:* Analysis sample for hire outcomes = hires\_obs (retained\_6mo observed); censored hires (started 2025-07 or later) are excluded from every hire regression and never imputed. *Source:* analysis/00\_load.py

**E01. Retention by quarter ATS**

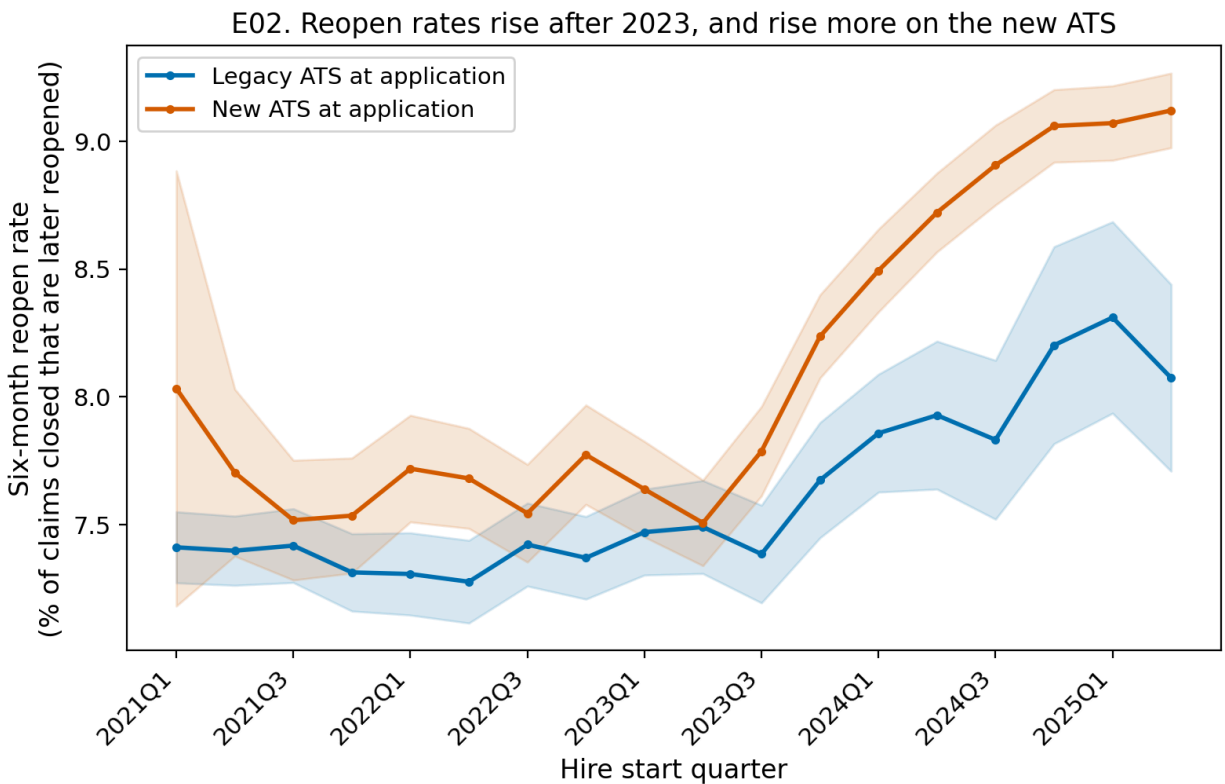
Sample: 41,677 permanent hires with an observed six-month outcome, start months 2021-01 to 2025-06 (later starts are censored).

Source: analysis/02\_retention\_trend.py

**E01b. Retention DiD: new ATS x post-AI-era interaction**

| term                                          | estimate | SE    | CI<br>low | CI<br>high | p     | n      | clusters |
|-----------------------------------------------|----------|-------|-----------|------------|-------|--------|----------|
| new ATS effect, pre-AI era (2021-22)          | 0.003    | 0.009 | -0.015    | 0.020      | 0.772 | 41,677 | 40       |
| DiD interaction: extra new-ATS effect, AI era | -0.023   | 0.010 | -0.042    | -0.004     | 0.018 | 41,677 | 40       |
| post-era gap, new vs legacy (sum of the two)  | -0.021   | 0.009 | -0.038    | -0.004     | 0.017 | 41,677 | 40       |

*Sample:* n=41677, 40 center clusters. *Notes:* Outcome: retained\_6mo (0/1, linear probability). FE: center, start\_month; post\_ai (application-month era) entered explicitly because start-month FE do not absorb it for hires who applied before and started after the cutoff. Controls: resume/license/experience/referral/employed/wage/unemployment. Cluster: center. Row 1 = new-ATS effect before the AI era; row 2 = the DiD interaction; row 3 = their sum, the new-vs-legacy gap in the AI era. *Source:* analysis/02\_retention\_trend.py

**E02. Reopen by quarter ATS**

Sample: 41,677 permanent hires with an observed six-month outcome.  
Source: analysis/02\_retention\_trend.py

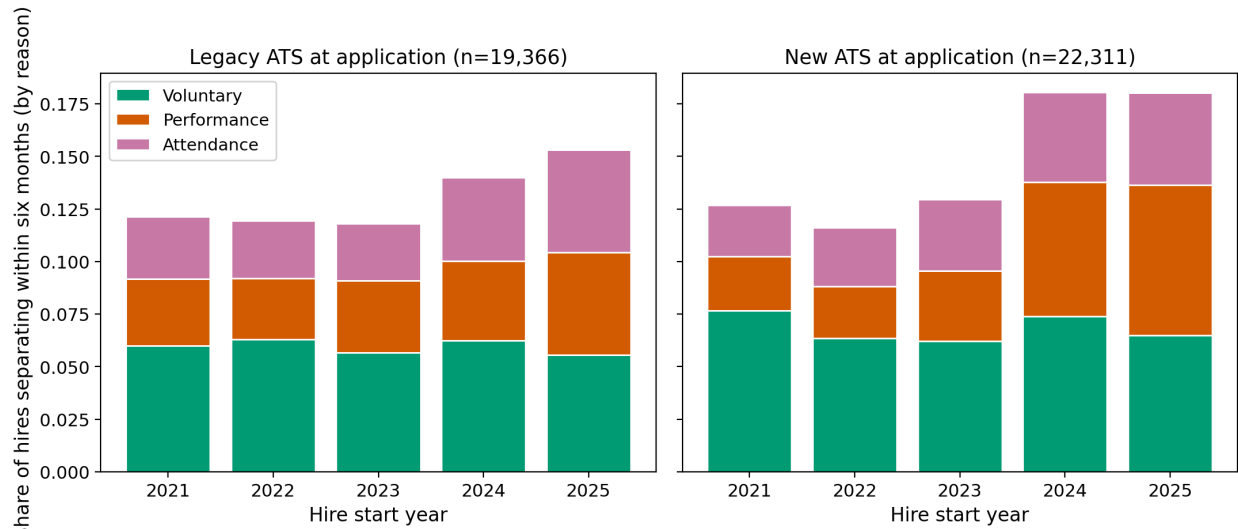
**E02b. Reopen-rate DiD: new ATS x post-AI-era interaction**

| term                                          | estimate | SE    | CI<br>low | CI<br>high | p     | n      | clusters |
|-----------------------------------------------|----------|-------|-----------|------------|-------|--------|----------|
| new ATS effect, pre-AI era (2021-22)          | 0.103    | 0.079 | -0.051    | 0.257      | 0.189 | 41,677 | 40       |
| DiD interaction: extra new-ATS effect, AI era | 0.145    | 0.091 | -0.033    | 0.323      | 0.111 | 41,677 | 40       |
| post-era gap, new vs legacy (sum of the two)  | 0.248    | 0.090 | 0.071     | 0.425      | 0.006 | 41,677 | 40       |

*Sample:* n=41677, 40 center clusters. *Notes:* Outcome: reopen\_rate\_6mo\_pct (percentage points). Same FE/controls/cluster as E01b. *Source:* analysis/02\_retention\_trend.py

### E03. Separation reason by year

E03. Separations rise after 2023 in every reason category, more on the new ATS



Sample: 41,677 permanent hires with an observed six-month outcome, by start year (2025 partial: only Jan-Jun starts have an outcome).

Source: analysis/02\_retention\_trend.py

#### E03b. Separation reason share by start year (%)

| year | voluntary | performance | attendance |
|------|-----------|-------------|------------|
| 2021 | 6.32      | 3.08        | 2.85       |
| 2022 | 6.32      | 2.73        | 2.76       |
| 2023 | 5.97      | 3.39        | 3.08       |
| 2024 | 7.16      | 5.86        | 4.21       |
| 2025 | 6.36      | 6.89        | 4.43       |

Sample: 41,677 hires with an observed six-month outcome. Source: analysis/02\_retention\_trend.py

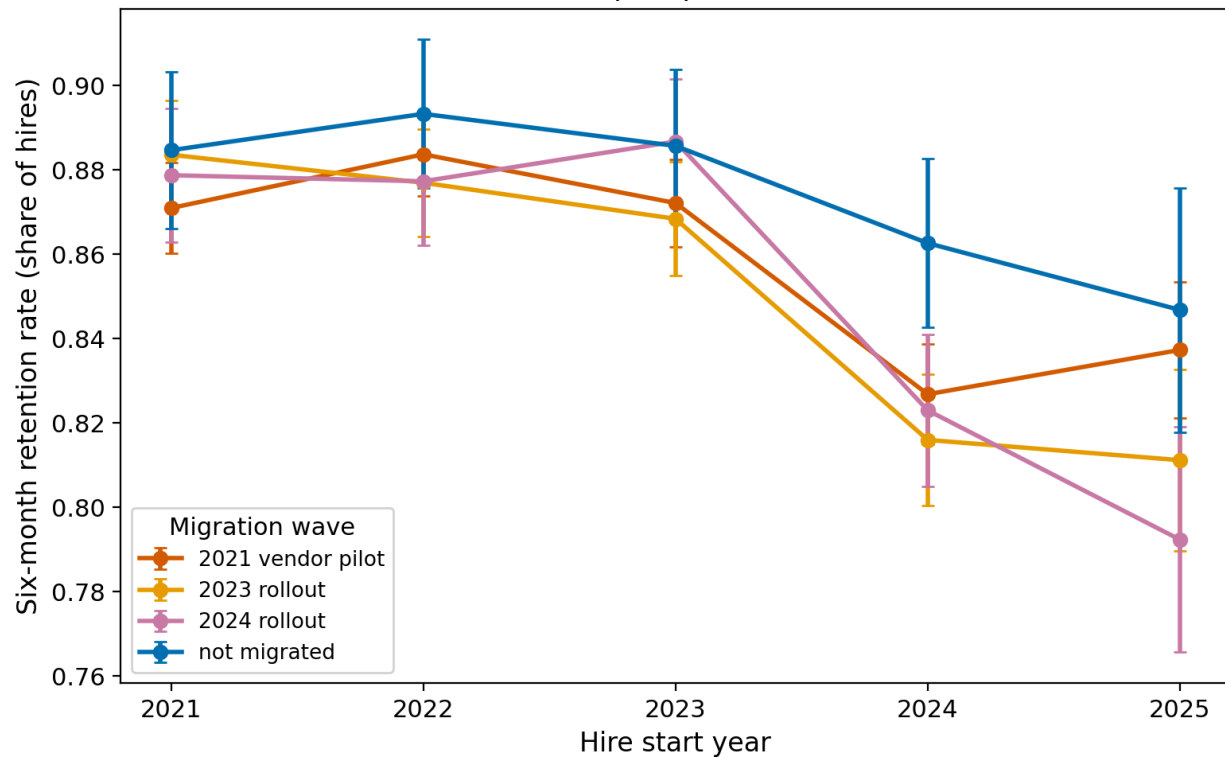
#### E03c. Separation-reason DiD: new ATS x post-AI-era interaction, by reason

|             | DiD estimate | CI low | CI high |
|-------------|--------------|--------|---------|
| voluntary   | 0.010        | -0.000 | 0.020   |
| performance | 0.010        | -0.001 | 0.021   |
| attendance  | 0.003        | -0.005 | 0.012   |

Sample: n=41677, 40 center clusters. Notes: Outcome: indicator for separating within six months for the stated reason. Same spec as E01b. Source: analysis/02\_retention\_trend.py

### E04. Retention by wave year

E04. The 2021 pilot centers -- the largest, best-resourced centers -- show the steepest post-2023 decline

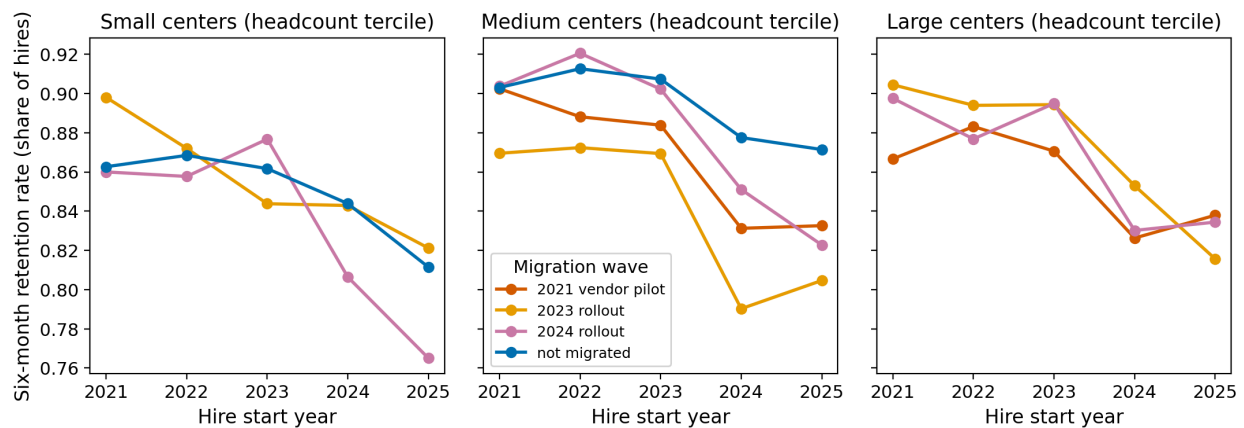


Sample: 41,677 permanent hires with an observed six-month outcome, grouped by their center's ATS migration wave.

Source: analysis/02\_retention\_trend.py

### E05. Retention by wave size tercile

E05. The pilot-wave decline shows up within every size tercile -- size alone does not explain it

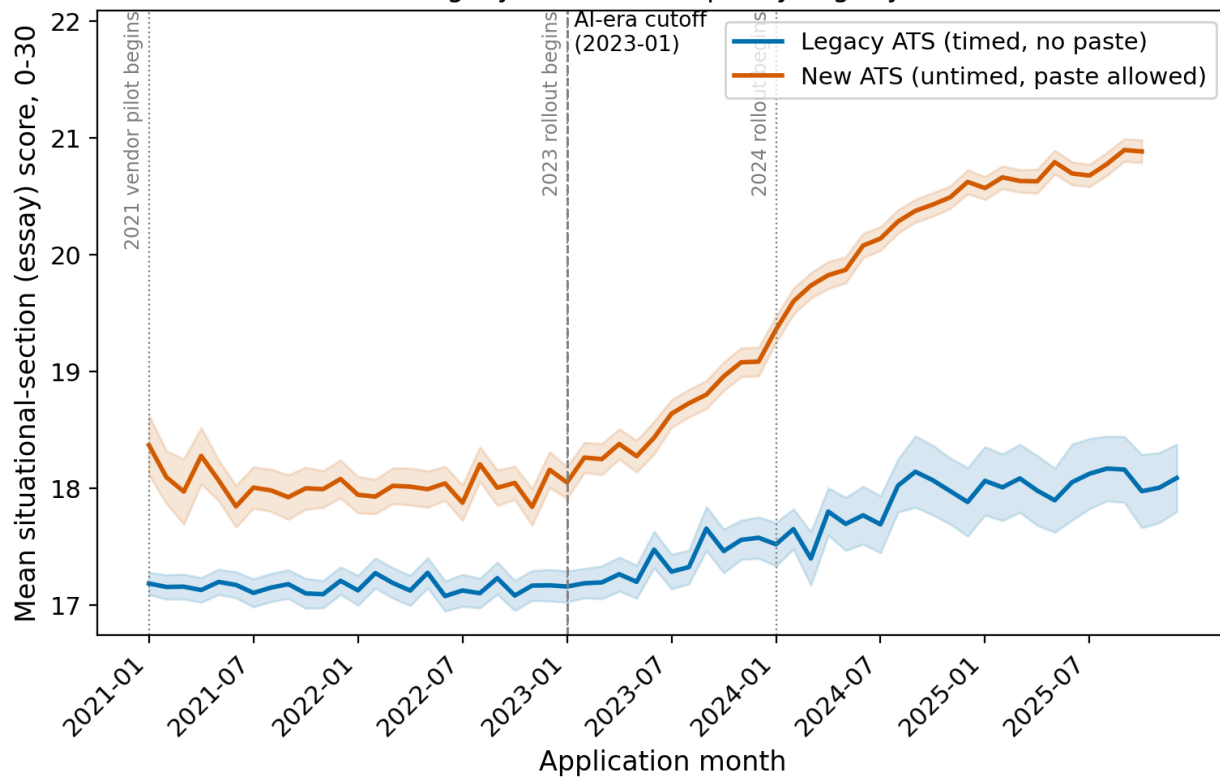


Sample: 41,677 permanent hires, split into center-headcount terciles.

Source: analysis/02\_retention\_trend.py

**E06. Essay score by ATS monthly**

E06. Essay scores climb on the new ATS from 2023;  
legacy scores drift up only slightly



Sample: 529,302 reviewed applications, 2021-01 to 2025-12, 40 centers. Shaded band is a 95% CI of the monthly mean.

Source: analysis/03\_pool\_quality.py

**E06b. Grid search for the AI-era break date in new-ATS essay scores**

| candidate month | pre mean | post mean | t     |
|-----------------|----------|-----------|-------|
| 2022-07         | 18.01    | 19.73     | 69.81 |
| 2022-08         | 18.01    | 19.75     | 73.13 |
| 2022-09         | 18.01    | 19.77     | 76.58 |
| 2022-10         | 18       | 19.8      | 80.37 |
| 2022-11         | 18.02    | 19.82     | 83.03 |
| 2022-12         | 18.01    | 19.85     | 86.22 |
| 2023-01         | 18.02    | 19.88     | 89.63 |
| 2023-02         | 18.01    | 19.9      | 92.93 |
| 2023-03         | 18.02    | 19.92     | 95.48 |
| 2023-04         | 18.02    | 19.96     | 99.17 |
| 2023-05         | 18.03    | 19.99     | 102.3 |
| 2023-06         | 18.05    | 20.03     | 105.4 |
| 2023-07         | 18.07    | 20.07     | 108.2 |
| 2023-08         | 18.08    | 20.1      | 111.2 |
| 2023-09         | 18.09    | 20.14     | 113.6 |
| 2023-10         | 18.12    | 20.18     | 115.9 |
| 2023-11         | 18.16    | 20.23     | 117.9 |
| 2023-12         | 18.19    | 20.27     | 119.7 |

Sample: Applications reviewed under the new ATS, 2021-01 to 2025-12. Notes: Best-fit break month by max t-stat: 2023-12. Configured cutoff in analysis/utls.py is 2023-01. Source: analysis/03\_pool\_quality.py

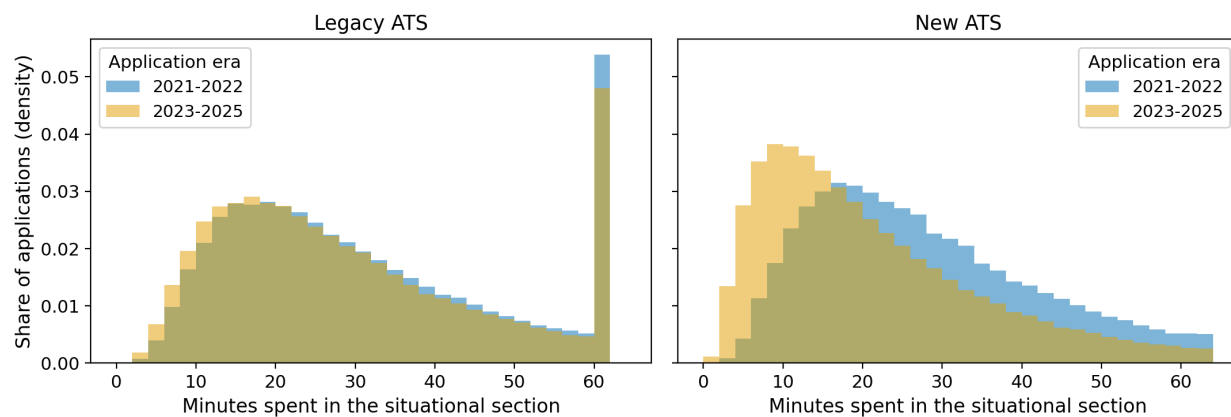
**E06c. Essay-score inflation: new ATS x post-AI-era interaction**

| term                                          | estimate | SE    | CI<br>low | CI<br>high | p      | n       | clusters |
|-----------------------------------------------|----------|-------|-----------|------------|--------|---------|----------|
| new ATS effect, pre-AI era (2021-22)          | 0.893    | 0.033 | 0.828     | 0.958      | <0.001 | 529,302 | 40       |
| DiD interaction: extra new-ATS effect, AI era | 0.787    | 0.108 | 0.575     | 1          | <0.001 | 529,302 | 40       |
| post-era gap, new vs legacy (sum of the two)  | 1.68     | 0.114 | 1.457     | 1.903      | <0.001 | 529,302 | 40       |

*Sample:* n=529302, 40 center clusters. *Notes:* Outcome: essay\_rubric\_score (0-30). Controls: resume/license/experience/referral/employed. FE: center, application\_month. Cluster: center. *Source:* analysis/03\_pool\_quality.py

**E07. Essay minutes distribution**

E07. Time spent on the essay collapses on the new ATS after 2023; legacy shifts only slightly



*Sample:* 529,302 reviewed applications, split by ATS and by application era.  
*Source:* analysis/03\_pool\_quality.py

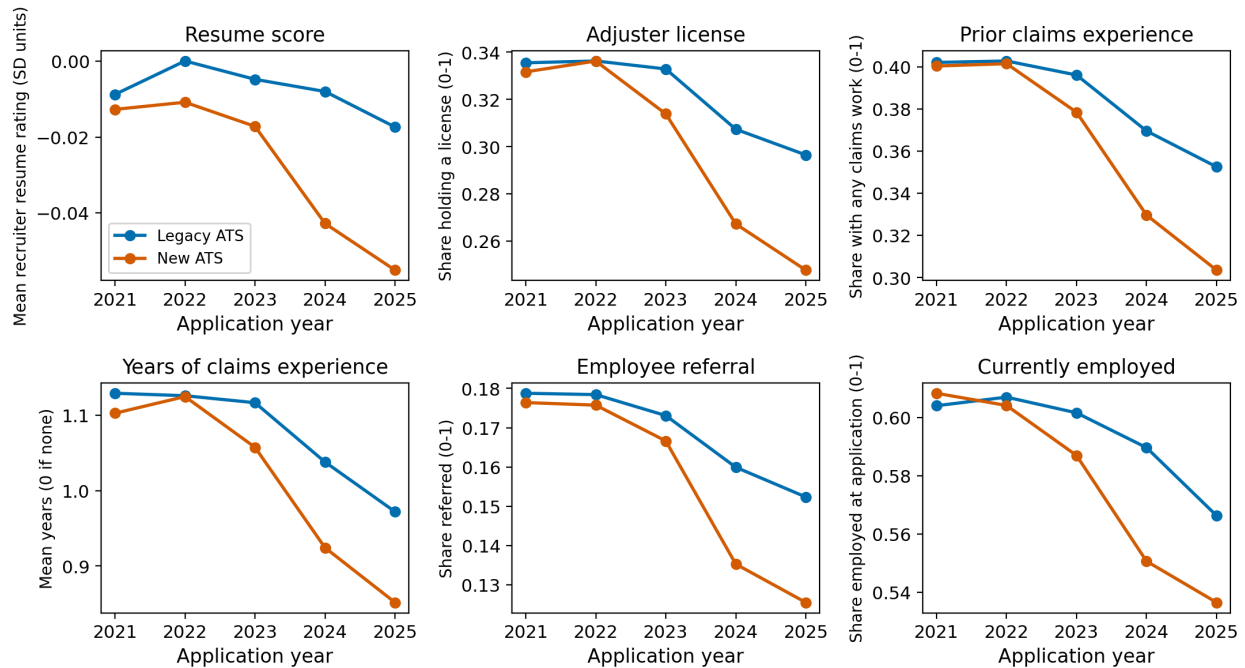
**E07b. Mean essay score and share of essays under 10 minutes, by year and ATS**

| year | ('legacy', 'essay') | ('legacy', 'fast10') | ('new', 'essay') | ('new', 'fast10') |
|------|---------------------|----------------------|------------------|-------------------|
| 2021 | 17.15               | 0.061                | 18.02            | 0.059             |
| 2022 | 17.16               | 0.062                | 18.01            | 0.063             |
| 2023 | 17.33               | 0.070                | 18.46            | 0.108             |
| 2024 | 17.75               | 0.094                | 19.87            | 0.222             |
| 2025 | 18.05               | 0.117                | 20.69            | 0.283             |

*Sample:* 529,302 reviewed applications. *Notes:* The legacy (timed, no-paste) system also drifts after 2023, more slowly: a timed field slows the leak, it does not stop it. *Source:* analysis/03\_pool\_quality.py

## E08. Objective attributes over time

E08. Objective applicant attributes drift slightly DOWN on the new ATS after 2023 -- the pool did not objectively improve while essay scores rose



Sample: 529,302 reviewed applications, yearly means by ATS at application.

Source: analysis/03\_pool\_quality.py

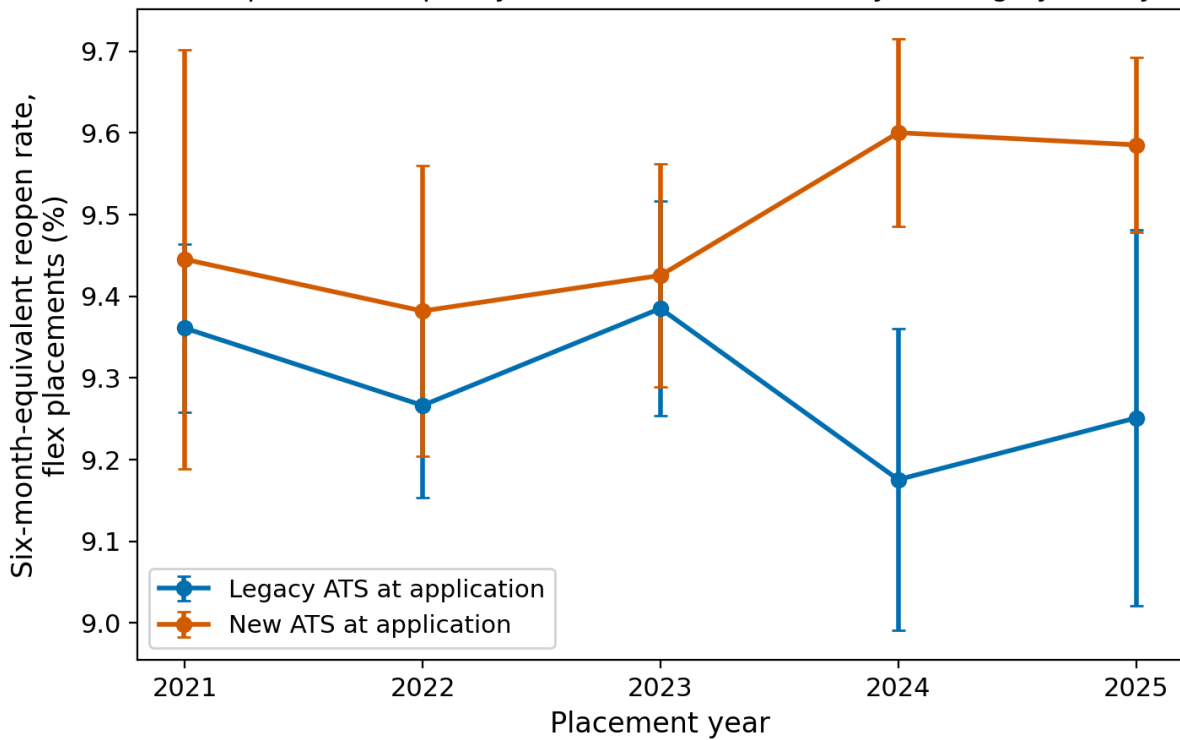
## E08b. Objective applicant attributes: new ATS x post-AI-era DiD

| outcome                 | term                                          | estimate | SE    | CI low | CI high | p      | n       | clusters |
|-------------------------|-----------------------------------------------|----------|-------|--------|---------|--------|---------|----------|
| resume_score            | new ATS effect, pre-AI era (2021-22)          | -0.003   | 0.006 | -0.015 | 0.008   | 0.559  | 529,302 | 40       |
| resume_score            | DiD interaction: extra new-ATS effect, AI era | -0.010   | 0.006 | -0.021 | 0.001   | 0.078  | 529,302 | 40       |
| resume_score            | post-era gap, new vs legacy (sum of the two)  | -0.013   | 0.007 | -0.027 | 0.000   | 0.057  | 529,302 | 40       |
| has_adjuster_license    | new ATS effect, pre-AI era (2021-22)          | -0.000   | 0.003 | -0.006 | 0.006   | 0.916  | 529,302 | 40       |
| has_adjuster_license    | DiD interaction: extra new-ATS effect, AI era | -0.026   | 0.004 | -0.035 | -0.018  | <0.001 | 529,302 | 40       |
| has_adjuster_license    | post-era gap, new vs legacy (sum of the two)  | -0.026   | 0.004 | -0.034 | -0.019  | <0.001 | 529,302 | 40       |
| prior_claims_experience | new ATS effect, pre-AI era (2021-22)          | 0.001    | 0.002 | -0.004 | 0.005   | 0.715  | 529,302 | 40       |
| prior_claims_experience | DiD interaction: extra new-ATS effect, AI era | -0.024   | 0.004 | -0.032 | -0.015  | <0.001 | 529,302 | 40       |
| prior_claims_experience | post-era gap, new vs legacy (sum of the two)  | -0.023   | 0.004 | -0.030 | -0.016  | <0.001 | 529,302 | 40       |
| prior_claims_years      | new ATS effect, pre-AI era (2021-22)          | -0.006   | 0.009 | -0.024 | 0.011   | 0.469  | 529,302 | 40       |
| prior_claims_years      | DiD interaction: extra new-ATS effect, AI era | -0.053   | 0.013 | -0.078 | -0.028  | <0.001 | 529,302 | 40       |
| prior_claims_years      | post-era gap, new vs legacy (sum of the two)  | -0.060   | 0.011 | -0.080 | -0.039  | <0.001 | 529,302 | 40       |
| referral                | new ATS effect, pre-AI era (2021-22)          | -0.004   | 0.002 | -0.007 | -0.000  | 0.043  | 529,302 | 40       |
| referral                | DiD interaction: extra new-ATS effect, AI era | -0.011   | 0.003 | -0.017 | -0.006  | <0.001 | 529,302 | 40       |
| referral                | post-era gap, new vs legacy (sum of the two)  | -0.015   | 0.003 | -0.021 | -0.009  | <0.001 | 529,302 | 40       |
| currently_employed      | new ATS effect, pre-AI era (2021-22)          | 0.003    | 0.003 | -0.003 | 0.009   | 0.342  | 529,302 | 40       |
| currently_employed      | DiD interaction: extra new-ATS effect, AI era | -0.021   | 0.003 | -0.027 | -0.015  | <0.001 | 529,302 | 40       |
| currently_employed      | post-era gap, new vs legacy (sum of the two)  | -0.018   | 0.004 | -0.025 | -0.011  | <0.001 | 529,302 | 40       |

Sample: n=529,302 reviewed applications, 40 center clusters. Notes: Each outcome regressed on treated\_app + treated\_app:post\_ai with center and application-month FE, cluster center. Compare with the essay-score DiD in E06c. Source: analysis/03\_pool\_quality.py

**E09. Flex reopen over time**

E09. Flex-placement quality (unselected on the essay) is roughly flat by ATS



Sample: 28,672 flex placements, 2021-01 to 2025-12. Error bars: 95% CI.  
Source: analysis/03\_pool\_quality.py

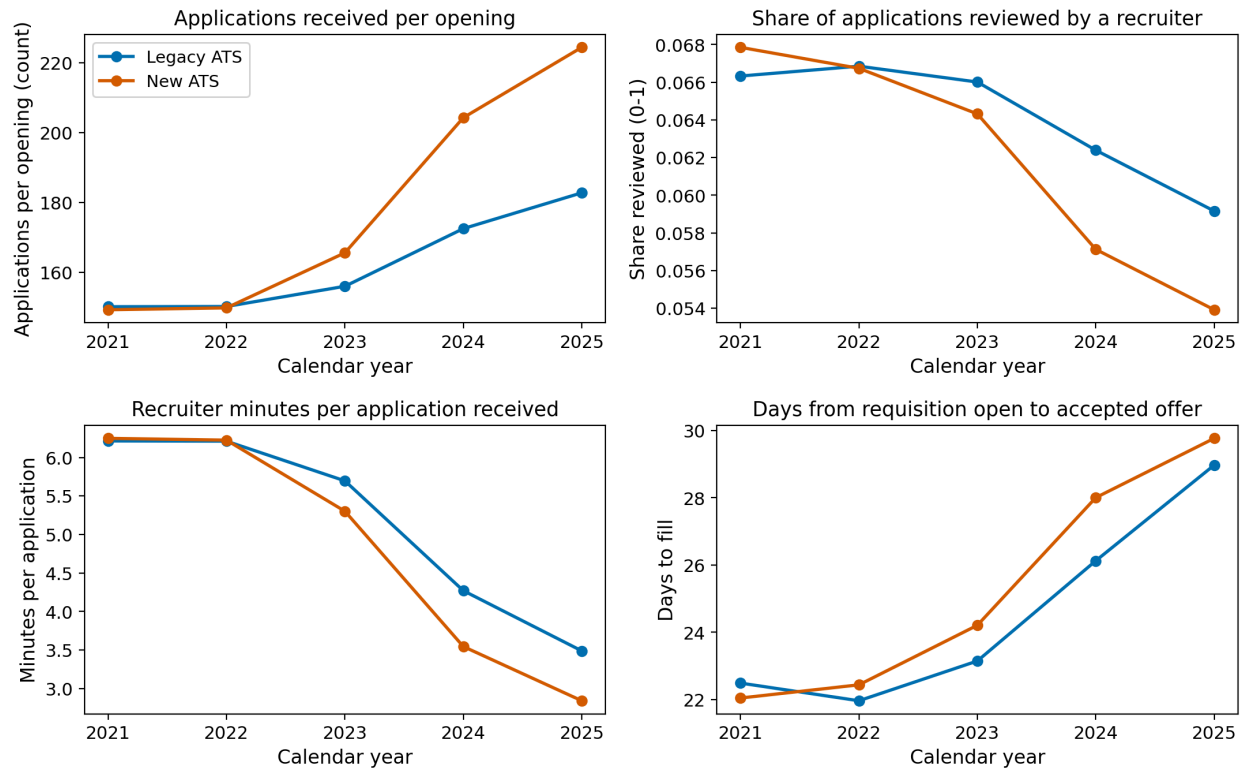
**E09b. Flex reopen rate: new ATS x post-AI-era interaction**

| term                                          | estimate | SE    | CI<br>low | CI<br>high | p     | n      | clusters |
|-----------------------------------------------|----------|-------|-----------|------------|-------|--------|----------|
| new ATS effect, pre-AI era (2021-22)          | 0.006    | 0.125 | -0.239    | 0.252      | 0.961 | 28,672 | 40       |
| DiD interaction: extra new-ATS effect, AI era | -0.075   | 0.136 | -0.342    | 0.192      | 0.581 | 28,672 | 40       |
| post-era gap, new vs legacy (sum of the two)  | -0.069   | 0.085 | -0.236    | 0.098      | 0.418 | 28,672 | 40       |

Sample: n=28672, 40 center clusters. Notes: Outcome: reopen\_rate\_pct. Controls: certification\_level FE. FE: center, placement\_month. Cluster: center. Source: analysis/03\_pool\_quality.py

## E10. Recruiter capacity panel

E10. Recruiter capacity has not kept pace with application volume:  
more applications per opening, less time per application, longer to fill



Sample: 2,400 center-months, 40 centers, 2021-01 to 2025-12, yearly means.  
Source: analysis/04\_recruiter\_capacity.py

### E10b. Recruiter-capacity metrics by year and ATS

| year | ats    | apps per opening | review share | recruiter minutes per application | days to fill |
|------|--------|------------------|--------------|-----------------------------------|--------------|
| 2021 | legacy | 150.2            | 0.066        | 6.21                              | 22.49        |
| 2021 | new    | 149.3            | 0.068        | 6.25                              | 22.04        |
| 2022 | legacy | 150.3            | 0.067        | 6.21                              | 21.96        |
| 2022 | new    | 149.9            | 0.067        | 6.22                              | 22.44        |
| 2023 | legacy | 156              | 0.066        | 5.7                               | 23.15        |
| 2023 | new    | 165.6            | 0.064        | 5.3                               | 24.21        |
| 2024 | legacy | 172.5            | 0.062        | 4.27                              | 26.13        |
| 2024 | new    | 204.3            | 0.057        | 3.54                              | 28           |
| 2025 | legacy | 182.8            | 0.059        | 3.49                              | 28.97        |
| 2025 | new    | 224.4            | 0.054        | 2.84                              | 29.77        |

Sample: 2,400 center-months, yearly averages by ATS in that center-month. Source: analysis/04\_recruiter\_capacity.py

**E10c. Drivers of days-to-fill**

|                                   | estimate | SE    | CI low | CI high | p      | n     | clusters |
|-----------------------------------|----------|-------|--------|---------|--------|-------|----------|
| Intercept                         | 21.37    | 3.894 | 13.74  | 29.01   | <0.001 | 2,400 | 40       |
| apps_per_opening                  | 0.028    | 0.011 | 0.006  | 0.049   | 0.011  | 2,400 | 40       |
| recruiter_minutes_per_application | -0.792   | 0.384 | -1.544 | -0.040  | 0.039  | 2,400 | 40       |
| local_unemployment_rate           | 0.537    | 0.275 | -0.002 | 1.075   | 0.051  | 2,400 | 40       |

*Sample:* n=2400, 40 center clusters. *Notes:* Outcome: days\_to\_fill. FE: center, month. Cluster: center. *Source:* analysis/04\_recruiter\_capacity.py

**E10d. Applications received per center-month: new ATS x AI-era DiD**

| term                                          | estimate | SE    | CI low | CI high | p      | n     | clusters |
|-----------------------------------------------|----------|-------|--------|---------|--------|-------|----------|
| new ATS effect, pre-AI era (2021-22)          | -478.3   | 91    | -656.7 | -300    | <0.001 | 2,400 | 40       |
| DiD interaction: extra new-ATS effect, AI era | 826.1    | 117.5 | 595.8  | 1,056   | <0.001 | 2,400 | 40       |
| post-era gap, new vs legacy (sum of the two)  | 347.8    | 126.8 | 99.3   | 596.3   | 0.010  | 2,400 | 40       |

*Sample:* n=2400 center-months, 40 clusters. *Notes:* FE: center, month. Cluster: center. *Source:* analysis/04\_recruiter\_capacity.py

**E10e. Recruiter-capacity metrics by year, all 40 centers pooled**

| year | apps per opening | review share | recruiter minutes per application | days to fill |
|------|------------------|--------------|-----------------------------------|--------------|
| 2021 | 150.1            | 0.067        | 6.22                              | 22.43        |
| 2022 | 150.1            | 0.067        | 6.22                              | 22.1         |
| 2023 | 160.6            | 0.065        | 5.51                              | 23.65        |
| 2024 | 195.9            | 0.059        | 3.73                              | 27.51        |
| 2025 | 217.1            | 0.055        | 2.95                              | 29.63        |

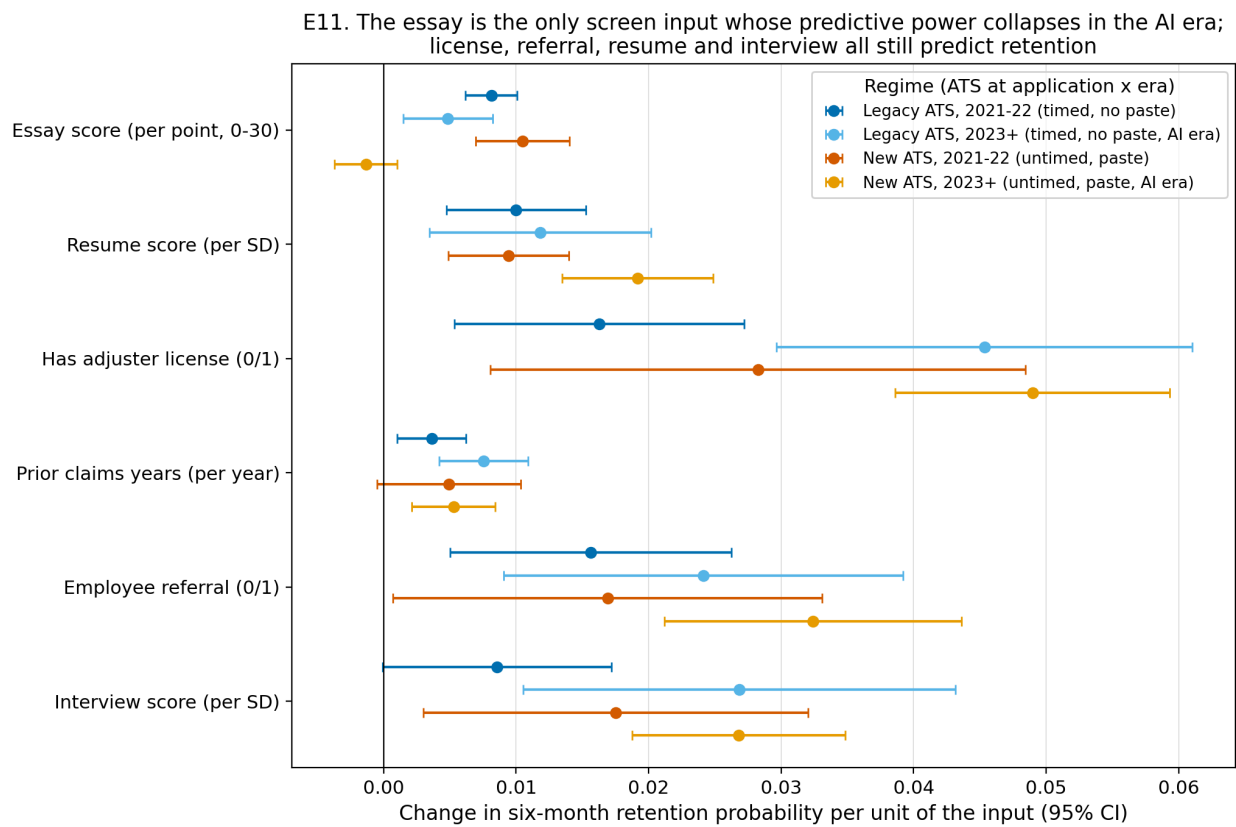
*Sample:* 2,400 center-months, yearly averages across all centers regardless of ATS. *Notes:* Pooled version of E10b; the narrative quotes these all-center figures. *Source:* analysis/04\_recruiter\_capacity.py

**E10f. Recruiter minutes per application vs applications per opening**

|                  | estimate | SE    | CI low | CI high | p      | n     | clusters |
|------------------|----------|-------|--------|---------|--------|-------|----------|
| Intercept        | 9.73     | 0.072 | 9.589  | 9.872   | <0.001 | 2,400 | 40       |
| apps_per_opening | -0.023   | 0.000 | -0.024 | -0.022  | <0.001 | 2,400 | 40       |

*Sample:* n=2400 center-months, 40 center clusters. *Notes:* Outcome: recruiter\_minutes\_per\_application. FE: center, month. Cluster: center. A negative coefficient means each extra applicant per opening buys less review time per application: recruiters spread a fixed capacity over a growing queue. *Source:* analysis/04\_recruiter\_capacity.py

## E11. Screen component coefficients by regime



Sample: 41,677 hires with an observed six-month outcome, split into four cells (legacy/new ATS x application before/after 2023-01); 4,589 censored hires excluded.  
Source: analysis/05\_screen\_components.py

## E11b. Predictors of six-month retention, by regime

| regime      | predictor            | estimate | SE    | CI low | CI high | n      | clusters |
|-------------|----------------------|----------|-------|--------|---------|--------|----------|
| legacy_pre  | essay_rubric_score   | 0.008    | 0.001 | 0.006  | 0.010   | 13,040 | 40       |
| legacy_pre  | resume_score         | 0.010    | 0.003 | 0.005  | 0.015   | 13,040 | 40       |
| legacy_pre  | has_adjuster_license | 0.016    | 0.006 | 0.005  | 0.027   | 13,040 | 40       |
| legacy_pre  | prior_claims_years   | 0.004    | 0.001 | 0.001  | 0.006   | 13,040 | 40       |
| legacy_pre  | referral             | 0.016    | 0.005 | 0.005  | 0.026   | 13,040 | 40       |
| legacy_pre  | interview_score      | 0.009    | 0.004 | -0.000 | 0.017   | 13,040 | 40       |
| legacy_post | essay_rubric_score   | 0.005    | 0.002 | 0.002  | 0.008   | 6,326  | 28       |
| legacy_post | resume_score         | 0.012    | 0.004 | 0.004  | 0.020   | 6,326  | 28       |
| legacy_post | has_adjuster_license | 0.045    | 0.008 | 0.030  | 0.061   | 6,326  | 28       |
| legacy_post | prior_claims_years   | 0.008    | 0.002 | 0.004  | 0.011   | 6,326  | 28       |
| legacy_post | referral             | 0.024    | 0.008 | 0.009  | 0.039   | 6,326  | 28       |
| legacy_post | interview_score      | 0.027    | 0.008 | 0.011  | 0.043   | 6,326  | 28       |
| new_pre     | essay_rubric_score   | 0.011    | 0.002 | 0.007  | 0.014   | 5,920  | 12       |
| new_pre     | resume_score         | 0.009    | 0.002 | 0.005  | 0.014   | 5,920  | 12       |
| new_pre     | has_adjuster_license | 0.028    | 0.010 | 0.008  | 0.048   | 5,920  | 12       |
| new_pre     | prior_claims_years   | 0.005    | 0.003 | -0.001 | 0.010   | 5,920  | 12       |
| new_pre     | referral             | 0.017    | 0.008 | 0.001  | 0.033   | 5,920  | 12       |
| new_pre     | interview_score      | 0.018    | 0.007 | 0.003  | 0.032   | 5,920  | 12       |
| new_post    | essay_rubric_score   | -0.001   | 0.001 | -0.004 | 0.001   | 16,391 | 33       |
| new_post    | resume_score         | 0.019    | 0.003 | 0.013  | 0.025   | 16,391 | 33       |
| new_post    | has_adjuster_license | 0.049    | 0.005 | 0.039  | 0.059   | 16,391 | 33       |
| new_post    | prior_claims_years   | 0.005    | 0.002 | 0.002  | 0.008   | 16,391 | 33       |
| new_post    | referral             | 0.032    | 0.006 | 0.021  | 0.044   | 16,391 | 33       |

| regime   | predictor       | estimate | SE    | CI low | CI high | n      | clusters |
|----------|-----------------|----------|-------|--------|---------|--------|----------|
| new_post | interview_score | 0.027    | 0.004 | 0.019  | 0.035   | 16,391 | 33       |

*Sample:* 41,677 hires with an observed outcome, split into four ATS x era cells. *Notes:* Outcome: retained\_6mo (linear probability). FE: center, start\_month. Cluster: center. Each cell is a separate regression on its subsample; all six predictors enter together. *Source:* analysis/05\_screen\_components.py

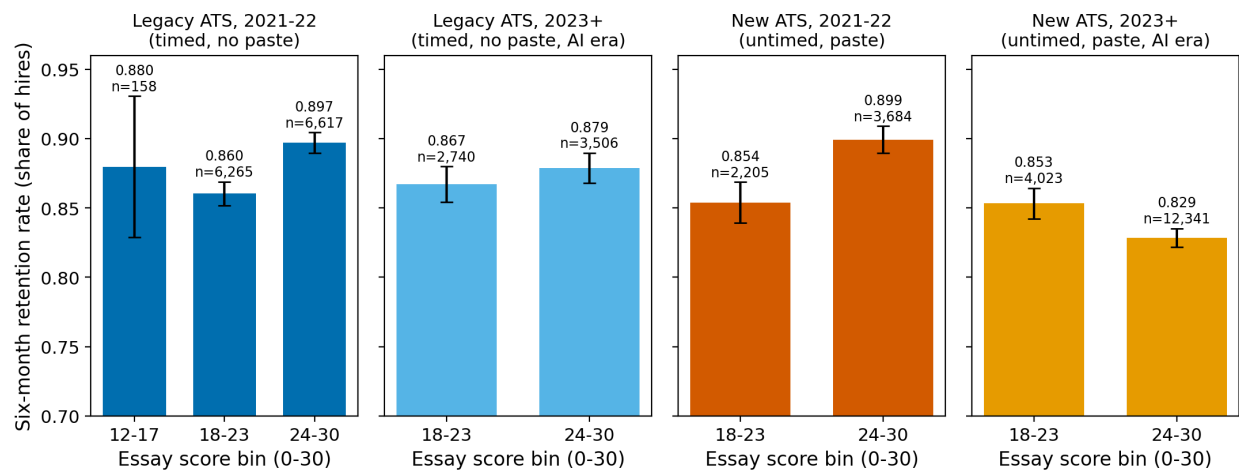
### E11c. How much weight recruiters place on the essay score, by regime

| regime      | essay weight in screen score | SE    | CI low | CI high | n       | r squared |
|-------------|------------------------------|-------|--------|---------|---------|-----------|
| legacy_pre  | 0.250                        | 0.000 | 0.250  | 0.251   | 135,320 | 0.862     |
| legacy_post | 0.250                        | 0.001 | 0.249  | 0.251   | 76,215  | 0.848     |
| new_pre     | 0.250                        | 0.000 | 0.249  | 0.250   | 62,232  | 0.861     |
| new_post    | 0.249                        | 0.000 | 0.249  | 0.250   | 255,535 | 0.821     |

*Sample:* All reviewed applications, split by regime. *Notes:* Outcome: recruiter\_screen\_score. FE: center, application\_month. Cluster: center. An R-squared near 1 means the screen score is a fixed formula of its inputs, not a judgment. *Source:* analysis/05\_screen\_components.py

### E12. Retention by essay bin regime

E12. A top-bin essay (24-30) used to predict better retention than an 18-23 essay; on the new ATS after 2023 it predicts slightly worse retention



*Sample:* 41,677 permanent hires with an observed six-month outcome, binned by essay score within four ATS x era cells; bins with under 100 hires not shown. Error bars: 95% CI. Y-axis starts at 0.70. *Source:* analysis/06\_essay\_validity.py

### E12b. Essay-score slope on six-month retention, by regime (hires)

| regime      | estimate | SE    | CI low | CI high | n cell |
|-------------|----------|-------|--------|---------|--------|
| legacy_pre  | 0.009    | 0.001 | 0.007  | 0.010   | 13,040 |
| legacy_post | 0.005    | 0.002 | 0.001  | 0.008   | 6,326  |
| new_pre     | 0.011    | 0.002 | 0.007  | 0.014   | 5,920  |
| new_post    | -0.003   | 0.001 | -0.005 | -0.000  | 16,391 |

*Sample:* n=41,677, 40 center clusters. *Notes:* Outcome: retained\_6mo. Controls: resume, license, prior years, referral, interview score. FE: center, start\_month. Cluster: center. Hires are selected on the essay, which attenuates every slope toward zero. *Source:* analysis/06\_essay\_validity.py

**E12c. Essay-score slope on six-month reopen rate, by regime (hires)**

| regime      | estimate | SE    | CI low | CI high | n cell |
|-------------|----------|-------|--------|---------|--------|
| legacy_pre  | -0.147   | 0.010 | -0.167 | -0.127  | 13,040 |
| legacy_post | -0.093   | 0.015 | -0.123 | -0.063  | 6,326  |
| new_pre     | -0.180   | 0.016 | -0.212 | -0.148  | 5,920  |
| new_post    | 0.035    | 0.010 | 0.016  | 0.054   | 16,391 |

*Sample:* n=41,677, 40 center clusters. *Notes:* Outcome: reopen\_rate\_6mo\_pct (percentage points).

Negative = higher essay score predicts fewer reopens. Same FE/cluster as E12b. *Source:* analysis/06\_essay\_validity.py

**E12d. Essay-score slope on reopen rate, by regime (flex placements -- NOT selected on the essay by Meridian)**

| regime      | estimate | SE    | CI low | CI high | n cell | weighted estimate | weighted SE |
|-------------|----------|-------|--------|---------|--------|-------------------|-------------|
| legacy_pre  | -0.179   | 0.010 | -0.199 | -0.160  | 9,079  | -0.177            | 0.010       |
| legacy_post | -0.122   | 0.012 | -0.146 | -0.098  | 5,541  | -0.118            | 0.013       |
| new_pre     | -0.207   | 0.019 | -0.244 | -0.169  | 2,346  | -0.212            | 0.021       |
| new_post    | -0.069   | 0.007 | -0.083 | -0.056  | 11,706 | -0.069            | 0.007       |

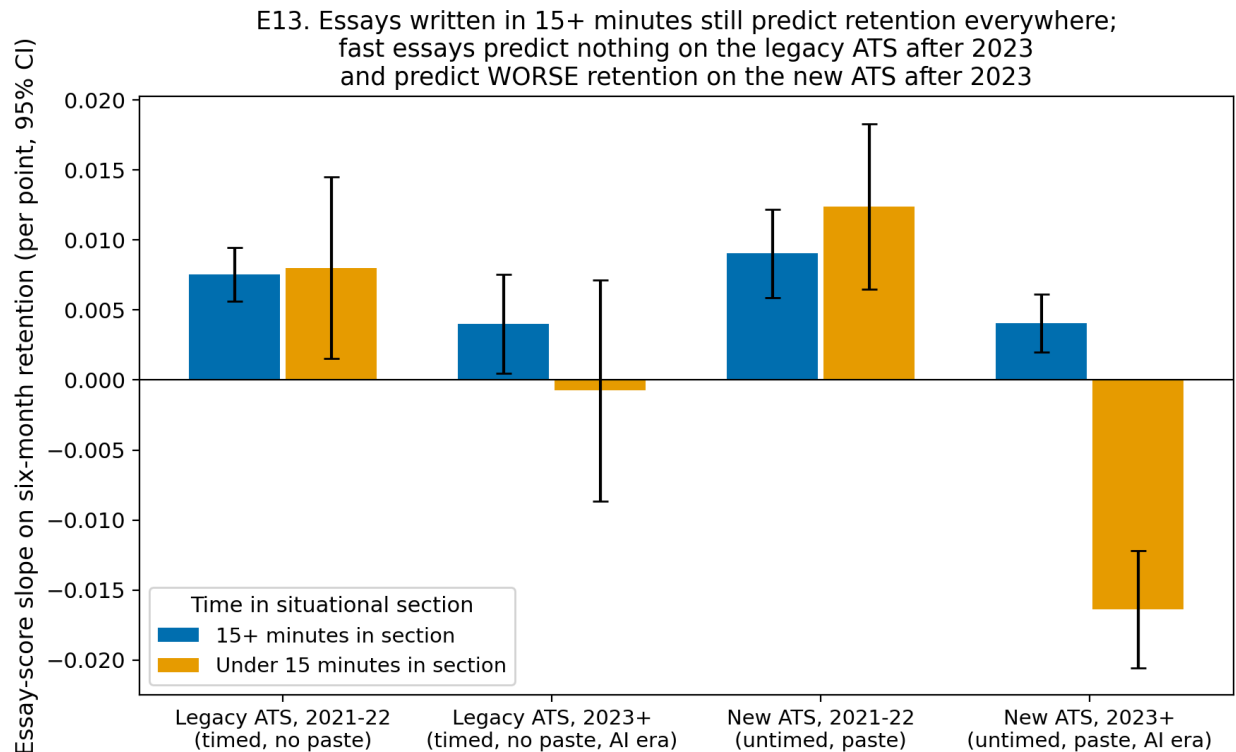
*Sample:* n=28,672 flex placements, 40 center clusters. *Notes:* Outcome: reopen\_rate\_pct. Controls: resume, license, prior years, referral, certification level. FE: center, placement\_month. Cluster: center. A MORE NEGATIVE slope means the essay is more informative. Weighted columns weight by claims\_closed. *Source:* analysis/06\_essay\_validity.py

**E12e. Spec C: essay slope on retention in the 12 months before vs after a center's own ATS migration (2023/24 waves) vs never-migrated centers over the same calendar window**

| group                                        | window           | estimate | SE    | CI low | CI high | n     |
|----------------------------------------------|------------------|----------|-------|--------|---------|-------|
| 2023/24-wave centers, own migration month    | 12 months before | 0.006    | 0.002 | 0.002  | 0.010   | 8,327 |
| 2023/24-wave centers, own migration month    | 12 months after  | -0.002   | 0.002 | -0.006 | 0.003   | 8,327 |
| never-migrated centers, pseudo-event 2023-09 | 12 months before | 0.007    | 0.003 | 0.002  | 0.012   | 2,359 |
| never-migrated centers, pseudo-event 2023-09 | 12 months after  | -0.003   | 0.002 | -0.007 | 0.001   | 2,359 |

*Sample:* 8,327 hires at 2023/2024-wave centers; 2,359 hires at never-migrated centers. *Notes:* Same controls/FE/cluster as E12b. The never-migrated slope also falls over this window (calendar 2022-09 to 2024-08), so the migration itself is not the whole story: the AI era degrades the timed section too, more slowly. *Source:* analysis/06\_essay\_validity.py

### E13. Essay slope by speed regime



Sample: 41,677 hires with an observed six-month outcome, split by ATS x era cell and by time spent in the situational section.

Source: analysis/06\_essay\_validity.py

### E13b. Essay-score slope on retention, by regime and by whether the applicant spent under 15 minutes in the situational section

| regime      | fast lt 15min | estimate | SE    | CI low | CI high | n cell |
|-------------|---------------|----------|-------|--------|---------|--------|
| legacy_pre  | 0             | 0.008    | 0.001 | 0.006  | 0.009   | 11,699 |
| legacy_pre  | 1             | 0.008    | 0.003 | 0.002  | 0.014   | 1,341  |
| legacy_post | 0             | 0.004    | 0.002 | 0.000  | 0.008   | 5,270  |
| legacy_post | 1             | -0.001   | 0.004 | -0.009 | 0.007   | 1,056  |
| new_pre     | 0             | 0.009    | 0.002 | 0.006  | 0.012   | 5,353  |
| new_pre     | 1             | 0.012    | 0.003 | 0.006  | 0.018   | 567    |
| new_post    | 0             | 0.004    | 0.001 | 0.002  | 0.006   | 9,876  |
| new_post    | 1             | -0.016   | 0.002 | -0.021 | -0.012  | 6,515  |

Sample: 41,677 hires with an observed outcome. Notes: Outcome: retained\_6mo. essay x cell x fast15 slopes with cell x fast15 main effects, center + start-month FE, cluster center. fast\_lt\_15min=1 means essay\_section\_minutes < 15. Source: analysis/06\_essay\_validity.py

**E13c. Essay score per minute spent in the section, by regime**

| regime      | estimate | SE    | n       |
|-------------|----------|-------|---------|
| legacy_pre  | 0.054    | 0.001 | 135,320 |
| legacy_post | 0.030    | 0.003 | 76,215  |
| new_pre     | 0.034    | 0.001 | 62,232  |
| new_post    | -0.021   | 0.001 | 255,535 |

*Sample:* All reviewed applications, by cell. *Notes:* essay\_rubric\_score ~ essay\_section\_minutes + center FE + application-month FE, cluster center. Positive = longer effort earns a higher score. *Source:* analysis/06\_essay\_validity.py

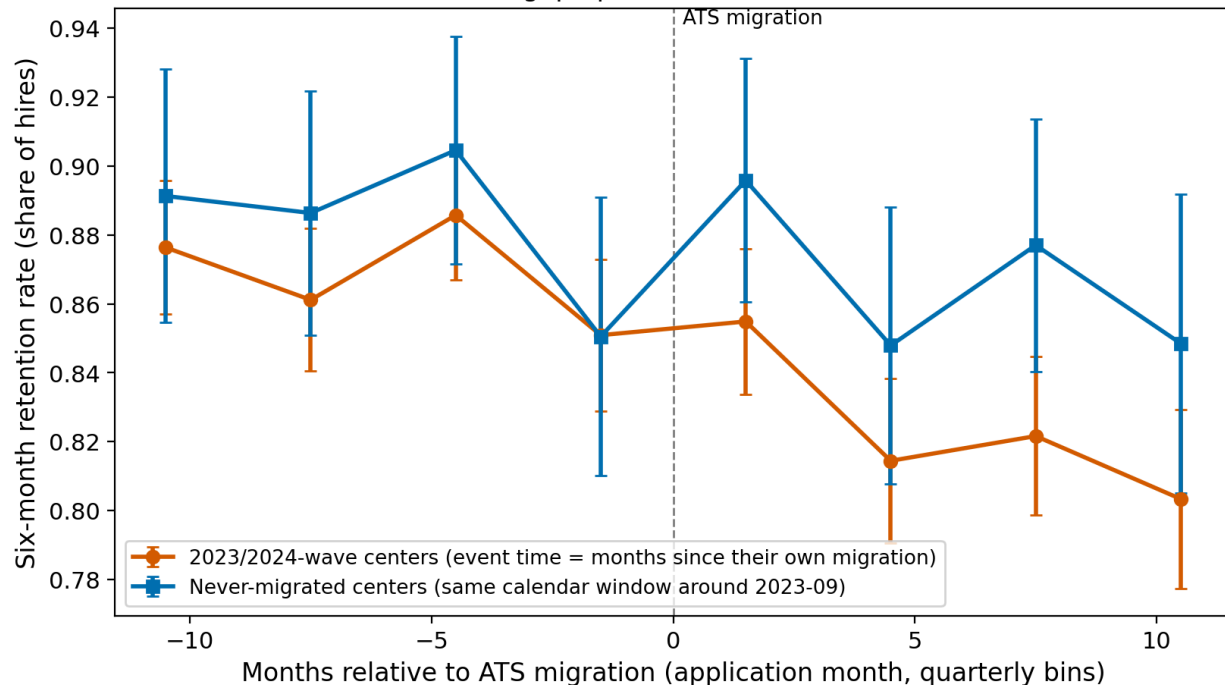
**E14. Placebo test: slope of other screen inputs on six-month retention, by regime**

| regime      | estimate | SE    | CI low | CI high | variable             |
|-------------|----------|-------|--------|---------|----------------------|
| legacy_pre  | 0.009    | 0.003 | 0.004  | 0.014   | resume_score         |
| legacy_post | 0.012    | 0.004 | 0.003  | 0.020   | resume_score         |
| new_pre     | 0.008    | 0.002 | 0.003  | 0.013   | resume_score         |
| new_post    | 0.023    | 0.003 | 0.017  | 0.028   | resume_score         |
| legacy_pre  | 0.020    | 0.006 | 0.009  | 0.030   | has_adjuster_license |
| legacy_post | 0.050    | 0.009 | 0.033  | 0.066   | has_adjuster_license |
| new_pre     | 0.033    | 0.010 | 0.014  | 0.052   | has_adjuster_license |
| new_post    | 0.060    | 0.005 | 0.050  | 0.071   | has_adjuster_license |
| legacy_pre  | 0.004    | 0.001 | 0.001  | 0.007   | prior_claims_years   |
| legacy_post | 0.009    | 0.002 | 0.005  | 0.012   | prior_claims_years   |
| new_pre     | 0.005    | 0.003 | 0.000  | 0.011   | prior_claims_years   |
| new_post    | 0.008    | 0.002 | 0.005  | 0.011   | prior_claims_years   |
| legacy_pre  | -0.005   | 0.004 | -0.013 | 0.004   | interview_score      |
| legacy_post | 0.019    | 0.008 | 0.003  | 0.035   | interview_score      |
| new_pre     | 0.001    | 0.007 | -0.013 | 0.014   | interview_score      |
| new_post    | 0.032    | 0.004 | 0.024  | 0.040   | interview_score      |

*Sample:* 41,677 hires with an observed outcome (interview\_score restricted to applicants who were interviewed). *Notes:* Univariate slopes (each input alone) with center + start-month FE, cluster center. If these are stable while the essay's slope collapses (E12b), the essay result is not an artifact of a center-wide shift in hiring quality or measurement. E11b gives the multivariate version. *Source:* analysis/06\_essay\_validity.py

## E15. Migration event study

E15. Migrating and never-migrated centers track each other before migration; the gap opens after the switch



Sample: 8,327 hires at 2023/2024-migration-wave centers (-12 to +11 months around each center's migration) and 2,359 hires at never-migrated centers (same window around the waves' median migration month). Error bars: 95% CI.  
Source: analysis/07\_identification.py

### E15b. Event-study means: retention before/after migration vs never-migrated

| period           | migrating centers | never migrated | gap never minus migrating |
|------------------|-------------------|----------------|---------------------------|
| 12 months before | 0.869             | 0.883          | 0.014                     |
| 12 months after  | 0.825             | 0.868          | 0.043                     |

Sample: 8,327 + 2,359 hires, event window +/-12 months. Source: analysis/07\_identification.py

### E15c. Selection on the essay into the hire sample, by regime

| regime4     | applicant<br>essay mean | applicant<br>essay sd | hire<br>essay mean | hire<br>essay sd | hire<br>rate<br>bin 18<br>23 | hire<br>rate<br>bin 24<br>30 | share<br>applicants 24 30 |
|-------------|-------------------------|-----------------------|--------------------|------------------|------------------------------|------------------------------|---------------------------|
| legacy_pre  | 17.16                   | 4.607                 | 23.65              | 2.957            | 0.120                        | 0.579                        | 0.084                     |
| legacy_post | 17.56                   | 4.74                  | 24                 | 2.975            | 0.097                        | 0.486                        | 0.105                     |
| new_pre     | 18.02                   | 4.591                 | 24.5               | 2.869            | 0.083                        | 0.501                        | 0.118                     |
| new_post    | 19.88                   | 4.832                 | 25.55              | 2.845            | 0.041                        | 0.274                        | 0.232                     |

Sample: All reviewed applications and all hires with an observed outcome. Notes: If the hire-sample essay SD were much smaller in one cell, a zero slope there could be a range-restriction artifact. It is not: the SD is similar in every cell. What changed is the share of applicants scoring 24-30 and the hire rate within that bin. Source: analysis/07\_identification.py

### E15d. Small-cluster inference: wild-cluster bootstrap on the headline essay-slope differences

| test                                                 | difference | cluster SE | cluster p | wild bootstrap p<br>299 | n      | clusters |
|------------------------------------------------------|------------|------------|-----------|-------------------------|--------|----------|
| hires: retention slope, new_post minus legacy_pre    | -0.011     | 0.001      | 0.000     | <0.001                  | 41,677 | 40       |
| hires: retention slope, legacy_post minus legacy_pre | -0.004     | 0.002      | 0.063     | 0.080                   | 41,677 | 40       |
| flex: reopen slope, new_post minus legacy_pre        | 0.110      | 0.013      | 0.000     | <0.001                  | 28,672 | 40       |
| flex: reopen slope, legacy_post minus legacy_pre     | 0.057      | 0.016      | 0.000     | 0.003                   | 28,672 | 40       |

*Sample:* Hires with an observed outcome (retention) and flex placements (reopen). *Notes:* H0: the essay slope in the named cell equals the legacy pre-2023 slope. Rademacher wild cluster bootstrap, 299 draws, 40 center clusters, restricted model imposes H0. A p below 0.05 means the difference survives small-cluster inference. *Source:* analysis/07\_identification.py

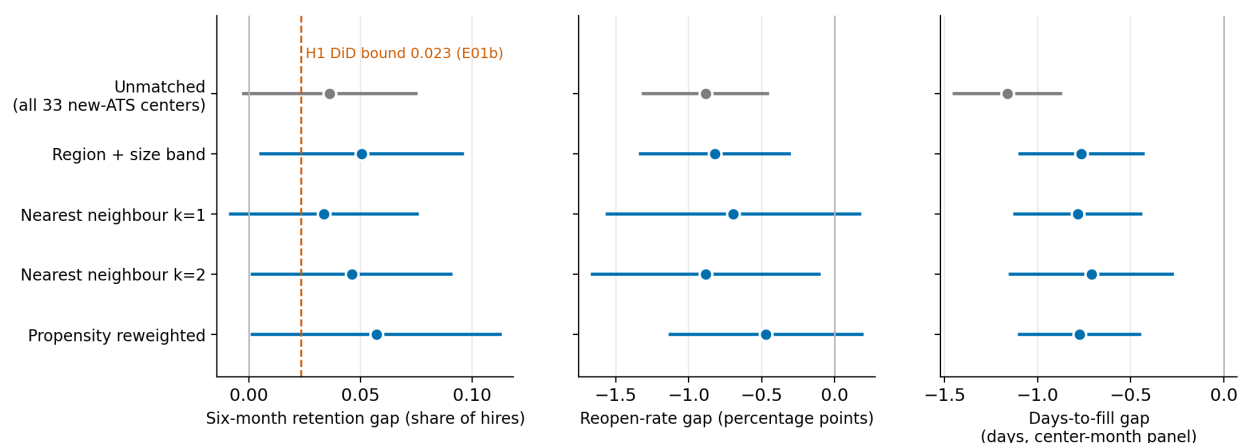
### E15e. Share of the legacy pre-2023 essay signal retained, by cell: hires sample vs flex sample

| regime      | hires share of legacy pre slope retained | flex share of legacy pre slope retained | range retained |
|-------------|------------------------------------------|-----------------------------------------|----------------|
| legacy_pre  | 1                                        | 1                                       | [100%, 100%]   |
| legacy_post | 0.547                                    | 0.681                                   | [55%, 68%]     |
| new_pre     | 1.232                                    | 1.153                                   | [115%, 123%]   |
| new_post    | 0.000                                    | 0.387                                   | [0%, 39%]      |

*Sample:* Slopes from E12b (hires) and E12d (flex). Negative hire slopes are clipped at 0% retained. *Notes:* The two samples bracket the truth: hires are selected on the essay (attenuated, lower bound on retained signal); flex are not. *Source:* analysis/07\_identification.py

### E15f. Matched legacy comparison

E15f. The legacy centers' retention advantage does not shrink once matched on size and region (unmatched 0.036, matched 0.046 [0.001, 0.091] with k=2 nearest neighbours), but 7 vs 5-19 centers is thin. All gaps are legacy minus new-ATS, 2024-25.



*Sample:* 1,737 hires at 7 never-migrated centers vs 11,440 new-ATS hires at 33 centers, starting 2024-01 or later, observed six-month outcome; days to fill from 168 + 750 center-months. Bars: 95% CI, SEs clustered by center (12-40 clusters; unreliable). *Source:* analysis/07\_identification.py

**E15f. Matched comparison: legacy-minus-new-ATS gaps in 2024-25, unmatched vs matched on size and region**

| approach | outcome             | unmatched gap | matched gap | matched CI low | matched CI high | wild boot p | matched centers | matched rows |
|----------|---------------------|---------------|-------------|----------------|-----------------|-------------|-----------------|--------------|
| band     | retained_6mo        | 0.036         | 0.051       | 0.005          | 0.096           | 0.074       | 9               | 2050         |
| band     | reopen_rate_6mo_pct | -0.884        | -0.821      | -1.331         | -0.311          | —           | 9               | 2050         |
| band     | days_to_fill        | -1.162        | -0.765      | -1.095         | -0.434          | —           | 9               | 198          |
| nn1      | retained_6mo        | 0.036         | 0.034       | -0.008         | 0.075           | —           | 5               | 1,342        |
| nn1      | reopen_rate_6mo_pct | -0.884        | -0.694      | -1.56          | 0.172           | —           | 5               | 1,342        |
| nn1      | days_to_fill        | -1.162        | -0.784      | -1.123         | -0.446          | —           | 5               | 118          |
| nn2      | retained_6mo        | 0.036         | 0.046       | 0.001          | 0.091           | —           | 7               | 1984         |
| nn2      | reopen_rate_6mo_pct | -0.884        | -0.884      | -1.661         | -0.107          | —           | 7               | 1984         |
| nn2      | days_to_fill        | -1.162        | -0.712      | -1.148         | -0.276          | —           | 7               | 166          |
| ps       | retained_6mo        | 0.036         | 0.057       | 0.002          | 0.113           | —           | 19              | 6,792        |
| ps       | reopen_rate_6mo_pct | -0.884        | -0.470      | -1.128         | 0.188           | —           | 19              | 6,792        |
| ps       | days_to_fill        | -1.162        | -0.775      | -1.098         | -0.453          | —           | 19              | 436          |

*Sample:* Legacy = 7 never-migrated centers, 1,737 hires starting 2024-01 or later with an observed six-month outcome (168 center-months for days to fill). Comparison = new-ATS-at-application hires starting 2024-01 or later (unmatched: 33 centers, 11,440 hires; matched: see n columns). *Notes:* Gap = coefficient on a legacy dummy in outcome ~ legacy + start-month (or month) FE, linear probability model for retention, SEs clustered by center; center FE are not identified for a between-center contrast. Approaches: band = comparison centers in the legacy regions (Mid-Atlantic, Midwest, Southeast) with size\_index in [0.511, 1.088] (the legacy centers' min..max, no margin), unweighted; nn1/nn2 = each legacy center matched with replacement to its 1 or 2 nearest new-ATS centers on size\_index within the same region, comparison centers weighted by times matched, spread evenly over their hires; ps = logit P(legacy | size\_index, region) on the centers in legacy regions, comparison weight p/(1-p). Matched comparison centers, as center(weight): band: C00(1), C01(1), C05(1), C07(1), C14(1), C19(1), C25(1), C27(1), C32(1); nn1: C05(1), C12(1), C19(1), C25(3), C27(1); nn2: C01(1), C05(1), C07(2), C12(1), C19(4), C25(4), C27(1); ps: C00(1.1), C01(0.081), C02(0.0015), C03(0.12), C05(0.42), C07(0.32), C11(0.013), C12(0.24), C13(0.1), C14(0.42), C19(0.91), C22(0.13), C24(0.0025), C25(2.4), C27(0.22), C28(0.014), C32(0.42), C35(0.056), C36(0.039). Unmatched 95% CIs, legacy n and per-approach detail are in E15f\_matched\_comparison\_full.csv. With 12-26 clusters the cluster-robust CIs are unreliable; the wild-cluster bootstrap p (299 Rademacher draws) is reported for the unweighted band retention gap only. The H1 DiD bound (E01b) is 0.0232. The '(b, matched)' row in E19 uses the nn2 retention and reopen gaps. *Source:* analysis/07\_identification.py

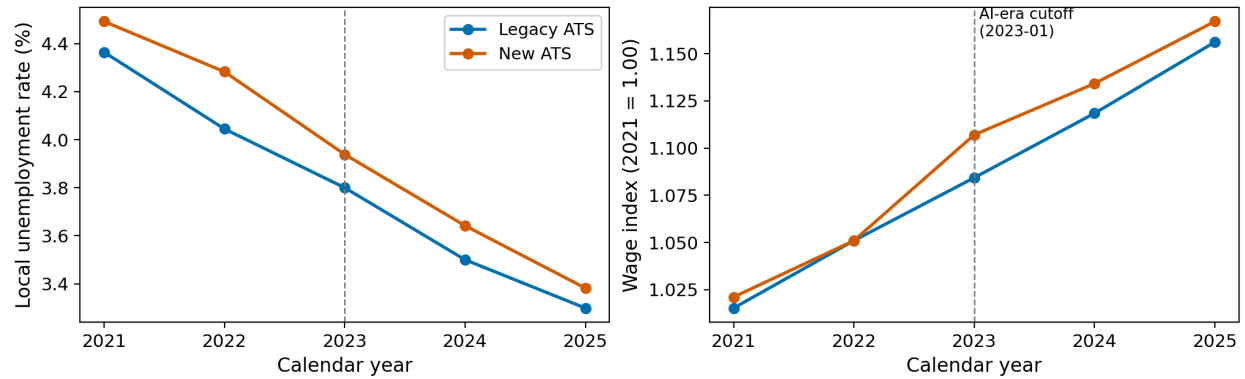
**E15g. Pre-2023 essay slope on retention: pilot centers vs never-migrated centers**

| centers                            | essay slope on retention | SE cluster center | CI low | CI high | n hires | clusters |
|------------------------------------|--------------------------|-------------------|--------|---------|---------|----------|
| 2021 vendor pilot (migrated 2021)  | 0.010                    | 0.001             | 0.007  | 0.012   | 8,012   | 12       |
| Never migrated (legacy throughout) | 0.008                    | 0.002             | 0.005  | 0.012   | 2,393   | 7        |

*Sample:* Hires applying before 2023-01 with an observed six-month outcome: 8,012 at pilot centers, 2,393 at never-migrated centers. *Notes:* retained\_6mo ~ essay\_rubric\_score + controls, center FE + start-month FE, SEs clustered by center. If migration had selected centers where the essay was more or less valid, these slopes would differ. *Source:* analysis/07\_identification.py

## E16. Labor market by ATS

E16. Local labor-market conditions move together regardless of ATS;  
they do not explain the new-vs-legacy divergence



Sample: 2,400 center-months, 40 centers, yearly means by ATS in that center-month; dashed line = 2023 AI-era cutoff.

Source: analysis/07\_identification.py

### E16b. Labor-market conditions by year and ATS

| year | ats    | local unemployment rate | wage index |
|------|--------|-------------------------|------------|
| 2021 | legacy | 4.364                   | 1.015      |
| 2021 | new    | 4.492                   | 1.021      |
| 2022 | legacy | 4.045                   | 1.051      |
| 2022 | new    | 4.283                   | 1.051      |
| 2023 | legacy | 3.8                     | 1.084      |
| 2023 | new    | 3.939                   | 1.107      |
| 2024 | legacy | 3.5                     | 1.118      |
| 2024 | new    | 3.642                   | 1.134      |
| 2025 | legacy | 3.299                   | 1.156      |
| 2025 | new    | 3.382                   | 1.167      |

Sample: 2,400 center-months. Source: analysis/07\_identification.py

**E17. Cost per retained employee: status quo vs counterfactuals where the essay still screened**

| scenario                                                                    | n      | r     | days<br>to fill | reopen % | C hire | C sep | C policy | C quality | CPR excl.<br>quality | CPR incl.<br>quality |
|-----------------------------------------------------------------------------|--------|-------|-----------------|----------|--------|-------|----------|-----------|----------------------|----------------------|
| Status quo: new ATS, hires starting 2024-25 (observed)                      | 11,440 | 0.820 | 28.93           | 8.918    | 18,699 | 2,540 | 0.000    | 17,531    | 23,377               | 40,908               |
| Status quo (alt): new ATS, all AI-era hires 2023-25 (observed)              | 16,391 | 0.834 | 28.93           | 8.592    | 18,699 | 2,540 | 0.000    | 16,888    | 22,916               | 39,804               |
| Counterfactual A: legacy centers, hires starting 2024-25 (observed)         | 2,487  | 0.857 | 27.26           | 7.985    | 18,334 | 2,540 | 0.000    | 15,695    | 21,821               | 37,516               |
| Counterfactual A': status quo + DiD retention gain, status-quo days to fill | —      | 0.843 | 28.93           | 7.985    | 18,699 | 2,540 | 0.000    | 15,695    | 22,665               | 38,360               |
| Counterfactual B: new ATS before the AI era (2021-22), costed at 2025 rates | 5,920  | 0.882 | 28.93           | 7.645    | 18,699 | 2,540 | 0.000    | 15,027    | 21,547               | 36,574               |

*Sample:* Hires with an observed six-month outcome, by scenario; all costed at 2025 average rates (E17a). CPR = cost per retained employee, \$. *Notes:* A = the seven never-migrated centers (different, smaller centers). A' = the status-quo cohort with retention raised by the H1 DiD gain (E01b) and reopen set to the legacy level. B = the same new-ATS centers before AI writing. C\_hire excludes recruiter screening time (E17b). *Source:* analysis/08\_cost\_worksheet.py

### E17a. Worksheet inputs: 2025 average rates across 40 centers vs the prompt's worked example

|                                 | 2025 average | prompt worked example |
|---------------------------------|--------------|-----------------------|
| sourcing_cost_per_opening       | 746.9        | 747                   |
| training_cost_per_hire          | 5,085        | 5,085                 |
| ramp_productivity_loss_per_hire | 6,540        | 6,540                 |
| vacancy_cost_per_day            | 218.7        | 219                   |
| separation_cost_per_exit        | 2,540        | 2,540                 |
| claims_per_adjuster_6mo         | 608.5        | 609                   |
| cost_per_reopened_claim         | 323          | 323                   |
| recruiter_hourly_rate           | 66.53        | —                     |

Sample: 480 center-months in 2025. Source: analysis/08\_cost\_worksheet.py

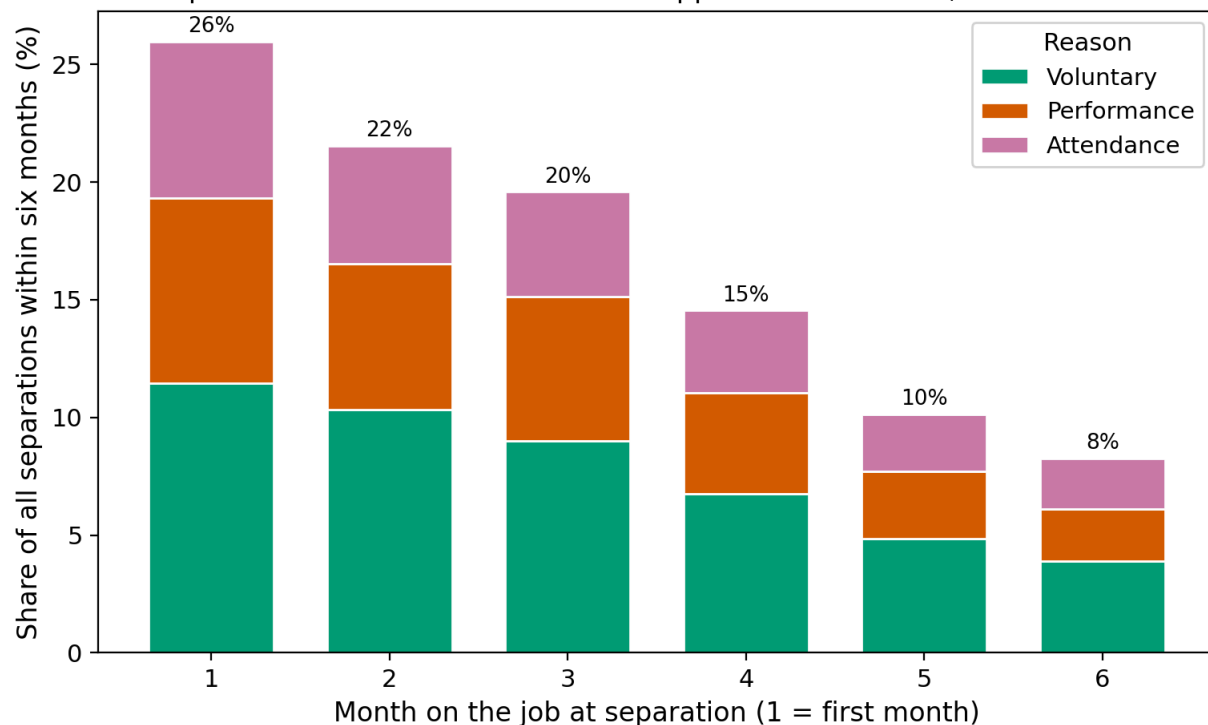
### E17b. Recruiter screening time per hire (not in the prompt's C\_hire; shown for option c)

|                                                               | value |
|---------------------------------------------------------------|-------|
| recruiter_minutes_per_application_2025                        | 2.954 |
| applications_received_per_hire_2025 (all centers, annualised) | 226   |
| recruiter_hourly_rate_2025                                    | 66.53 |
| recruiter_screening_cost_per_hire_\$                          | 740.4 |
| as_share_of_C_hire_status_quo                                 | 0.040 |

Sample: 2025 center-months and 2025 hires (H1 x 2). Source: analysis/08\_cost\_worksheet.py

### E18. Separation hazard by month

E18. Separations are front-loaded: 47% happen in months 1-2, 33% in months 4-6



Sample: 5,796 hires who separated within six months, out of 41,677 with an observed outcome (all regimes).

Source: analysis/08\_cost\_worksheet.py

**E18b. Separations by month on the job and reason (counts)**

| <b>months to sep</b> | <b>attendance</b> | <b>performance</b> | <b>voluntary</b> | <b>total</b> |
|----------------------|-------------------|--------------------|------------------|--------------|
| 1                    | 385               | 457                | 663              | 1,505        |
| 2                    | 291               | 359                | 598              | 1,248        |
| 3                    | 258               | 355                | 522              | 1,135        |
| 4                    | 203               | 249                | 391              | 843          |
| 5                    | 140               | 166                | 281              | 587          |
| 6                    | 124               | 129                | 225              | 478          |

*Sample:* 5,796 hires who separated within six months. *Source:* analysis/08\_cost\_worksheet.py

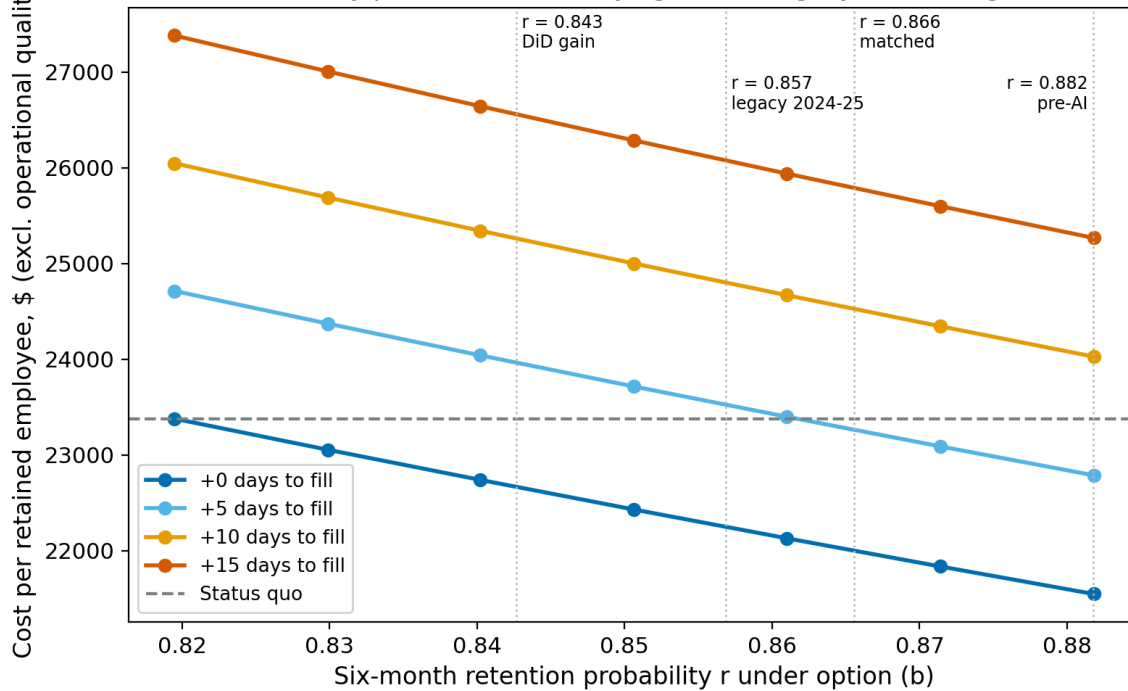
**E19. Cost per retained employee by option, vs status quo**

| scenario                                                                                                                                                                                | n      | r     | days<br>to<br>fill | reopen<br>% | C<br>hire | C<br>sep | C<br>policy | C<br>quality | CPR<br>excl.<br>quality | CPR<br>incl.<br>quality | delta vs<br>status<br>quo<br>excl.<br>quality | annual<br>delta<br>excl.<br>quality<br>\$M | delta vs<br>status<br>quo incl.<br>quality | annual<br>delta<br>incl.<br>quality<br>\$M |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|--------------------|-------------|-----------|----------|-------------|--------------|-------------------------|-------------------------|-----------------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|
| Status quo: new ATS, hires starting 2024-25 (observed)                                                                                                                                  | 11,440 | 0.820 | 28.93              | 8.918       | 18,699    | 2,540    | 0.000       | 17,531       | 23,377                  | 40,908                  | 0.000                                         | 0.000                                      | 0.000                                      | 0.000                                      |
| (a) Do nothing: status quo continues                                                                                                                                                    | 11,440 | 0.820 | 28.93              | 8.918       | 18,699    | 2,540    | 0.000       | 17,531       | 23,377                  | 40,908                  | 0.000                                         | 0.000                                      | 0.000                                      | 0.000                                      |
| (a') Do nothing, 2024->25 retention change (+0.000) continues one more year                                                                                                             | —      | 0.820 | 28.93              | 8.918       | 18,699    | 2,540    | 0.000       | 17,531       | 23,354                  | 40,885                  | -22.88                                        | -0.160                                     | -22.88                                     | -0.160                                     |
| (b, low) Applicants bear it: retention gain = H1 DiD only; reopen = legacy                                                                                                              | —      | 0.843 | 28.93              | 7.985       | 18,699    | 2,540    | 0.000       | 15,695       | 22,665                  | 38,360                  | -712.6                                        | -4.87                                      | -2,548                                     | -17.4                                      |
| (b) Applicants bear it: timed/no-paste section; r and reopen = legacy centers 2024-25; status-quo days to fill                                                                          | 2,487  | 0.857 | 28.93              | 7.985       | 18,699    | 2,540    | 0.000       | 15,695       | 22,247                  | 37,942                  | -1,130                                        | -7.72                                      | -2,966                                     | -20.26                                     |
| (b, high) Applicants bear it: legacy r, reopen AND legacy days to fill                                                                                                                  | 2,487  | 0.857 | 27.26              | 7.985       | 18,334    | 2,540    | 0.000       | 15,695       | 21,821                  | 37,516                  | -1,556                                        | -10.63                                     | -3,392                                     | -23.17                                     |
| (b, matched) Applicants bear it: r and reopen = status quo + legacy gap matched on size and region (nearest neighbour, k=2, on size_index within region, E15f); status-quo days to fill | —      | 0.866 | 28.93              | 8.035       | 18,699    | 2,540    | 0.000       | 15,794       | 21,998                  | 37,792                  | -1,379                                        | -9.42                                      | -3,116                                     | -21.28                                     |
| (c) Meridian bears it: structured interview; ASSUMED r recovers 50% of the legacy gap, +15 recruiter min per reviewed application (\$204/hire), +3 days to fill                         | —      | 0.838 | 31.93              | 8.452       | 19,559    | 2,540    | 0.000       | 16,613       | 23,826                  | 40,439                  | 448.5                                         | 3.06                                       | -469.4                                     | -3.21                                      |
| (d) Bear it after hiring: \$500 retention bonus at six months; ASSUMED r recovers 17% of the legacy gap (separations are front-loaded, E18)                                             | —      | 0.826 | 28.93              | 8.918       | 18,699    | 2,540    | 500         | 17,531       | 23,682                  | 41,213                  | 304.5                                         | 2.08                                       | 304.5                                      | 2.08                                       |

*Sample:* Status quo, (a) and (b) are observed cohorts; (a'), (b low/high/matched), (c) and (d) rest on the stated assumptions in the row label. Annual figures use 6,830 retained seats a year (8,334 new-ATS hires x r, 2025 H1 pace x 2). *Notes:* Negative delta = cheaper than status quo. (c) and (d) are not estimated from Meridian data because Meridian has not run them; their retention effects are assumptions and E20 shows how far they can move before the ranking changes. *Source:* analysis/08\_cost\_worksheet.py

**E20. Sensitivity option b**

E20. Option (b) stays cheaper than the status quo unless a timed section adds roughly a week of vacancy per hire without buying the full legacy retention gain



Sample: Worksheet evaluated over a grid of  $r$  (status quo to pre-AI level) and extra days to fill; other inputs at 2025 averages; quality excluded.

Source: analysis/08\_cost\_worksheet.py

**E20b. Sensitivity table: cost per retained employee under option (b), \$ excl. quality, by  $r$  (rows) and extra days to fill (columns)**

| $r$   | 0      | 5      | 10     | 15     |
|-------|--------|--------|--------|--------|
| 0.820 | 23,377 | 24,711 | 26,045 | 27,379 |
| 0.830 | 23,052 | 24,370 | 25,687 | 27,004 |
| 0.840 | 22,739 | 24,040 | 25,341 | 26,642 |
| 0.851 | 22,430 | 23,715 | 25,000 | 26,285 |
| 0.861 | 22,128 | 23,398 | 24,667 | 25,937 |
| 0.871 | 21,834 | 23,088 | 24,343 | 25,597 |
| 0.882 | 21,546 | 22,786 | 24,026 | 25,265 |

Sample: 2025-average cost rates. Source: analysis/08\_cost\_worksheet.py

### E20c. Break-even: extra vacancy days that erase option (b)'s saving; recovery share option (c) needs

| r<br>assumed | label                                             | breakeven<br>extra days<br>excl.<br>quality | breakeven<br>extra days<br>incl.<br>quality | option c<br>recovery<br>share<br>needed to<br>break<br>even excl.<br>quality | at extra<br>recruiter<br>cost per<br>hire \$ | at<br>extra<br>days<br>to fill |
|--------------|---------------------------------------------------|---------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------|--------------------------------|
| 0.843        | DiD gain only                                     | 2.746                                       | 9.82                                        | 0.890                                                                        | 204.1                                        | 3                              |
| 0.857        | legacy 2024-25 observed                           | 4.429                                       | 11.62                                       | —                                                                            | —                                            | —                              |
| 0.866        | legacy gap matched on size and region (nn2, E15f) | 5.459                                       | 12.72                                       | —                                                                            | —                                            | —                              |
| 0.882        | pre-AI level                                      | 7.38                                        | 14.78                                       | —                                                                            | —                                            | —                              |

*Sample:* 2025-average rates; status quo from E19 row 0. *Notes:* Each extra day to fill costs vacancy\_cost\_per\_day / r per retained employee. *Source:* analysis/08\_cost\_worksheet.py